宣言与五篇 · THE DECLARATION AND FIVE ESSAYS

时间主权宣言

完整版

The Sovereignty of Time: A Declaration · Complete Edition

Titan

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宣言四层与"时间盗窃"系列五篇，中文在前，英文在后。十二个规范文本的 SHA-256 见版权页；本文件的 SHA-256 记录于比特币区块链（OpenTimestamps），原文托管于 SSRN。

The four-layer declaration and the five essays of "The Theft of Time," Chinese first, English second. The colophon carries the SHA-256 of the twelve canonical files; the SHA-256 of this file is timestamped on the Bitcoin blockchain (OpenTimestamps) and the text is hosted on SSRN.

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一句话

法币偷走你的过去，算法偷走你的现在，AI 偷走你的未来；时间主权，就是把三者夺回来。

一段话

时间是唯一不可再生的资源。三套系统先后学会了偷它：法币稀释你用时间换来的储蓄，算法切碎你当下的注意力，AI 正在拿走你对未来时间的定价权——而它们都不需要你同意。AI 之后，生产不再稀缺，稀缺的是分配。时间主权不是逃离，而是把存在本身记录在一个不能被稀释的账本上，让每一份被机器放大的产出，都向人的时间纳税。这不是唯一的解，但它是一个可以被验证的解。

一页版

一

时间是唯一不可再生的资源。钱可以再赚，信息可以复制，能量可以转换，时间只会单向流走。所以一切主权的底层，是对时间的主权。你的一生，是一笔正在不可逆支出的时间。问题只有一个：它流向了谁。

二

三套系统，先后学会了偷它。

法币偷走你的过去。储蓄是你已经付出的时间的凝结，每一次增发都在稀释它。通胀不是经济现象，它是针对储蓄者的时间盗窃。

算法偷走你的现在。注意力是时间在当下的形态，而推荐系统的生意，就是把它切碎、拍卖、再切碎。

AI 偷走你的未来。专业知识曾是你对未来时间的定价权；当认知可以被无限复制，定价权就不再在你手里。

三者有一个共同点：它们都不需要你同意。

三

宇宙的铁律是熵增，但生命从来不是靠对抗它活下来的。生命是开放系统：消耗一个能量梯度，换取局部的秩序。熵增是宇宙的税，秩序是你用梯度买来的东西。所以时间领主的问题从来不是“如何抗争”，而是：你的梯度是什么，它是否正在被别人收走。你的身体、你的注意力、你的部落，是你仅有的三个梯度。

四

旧的分配机制是劳动：你出时间，换回份额。当机器出时间，这条链就断了。AI 之后，生产不再稀缺，稀缺的是分配。机器放大一切产出，却没有任何机制规定放大的部分归谁。一个文明可以同时拥有无限的产出和无处安放的人——这不是技术问题，是账本问题。捕获价值的机制，不能同时是分配价值的机制。

五

换一种货币，不够。比特币改变了货币的发行权，模因币取消了发行的门槛，稳定币干脆把法币搬上了链——没有一个逃出法币：只要涨跌仍由资金决定，价值就仍以法币计价，参与者就仍在赌场的规则里。它们只换了账本上的单位，没有换账本记的东西。

已经有人把另一种思路写成了协议，在几条链上实验了近四年：把存在本身记录为不可稀释的凭证——存在不是身份，不是资产，是你在时间中出现过这个事实；让注意力和创造力在其上定价；让每一份被机器放大的产出，向人的时间纳税。它暴露了问题，也证明了可能。它不是唯一的解，未必是最好的解，但它是一个可以被验证的解。我更希望看到更多人提出自己的分配协议。

六

终局不是逃离，是延续。肉体的终点由生物学决定，影响力的终点由账本决定。把你的思想、决策和作品刻进永不关闭的档案，让它们在你之后继续被读取、被执行、被继承。如果时间是货币，死亡就是最终的破产——而代码第一次给了我们重组的机会。

七

你无法选择时间是否流走，只能选择它记在谁的账上。

从今天起，记在你自己的账上。

时间主权宣言

The Sovereignty of Time: A Declaration

Titan · v1.1 · 2026 年 11 月

一、时间

时间是唯一不可再生的资源。钱可以再赚，信息可以复制，能量可以转换，时间只会单向流走。这不是修辞，是物理：熵增给了时间一个方向，而你恰好活在这个方向上。

所以一切主权的底层，是对时间的主权。领土主权、货币主权、数据主权，说到底都在回答同一个问题：谁有权支配一段时间里发生的事。对一个人来说，这个问题更简单，也更残酷——你的一生，是一笔正在不可逆支出的时间，而你从来没有看过这笔账。

这份宣言只做一件事：把这笔账摊开。看清它流向了谁，为什么你没有同意，以及怎样把它记回你自己的名下。

二、三重盗窃

三套系统，先后学会了偷时间。

我说“偷”，意思很具体：未经同意的转移。不是抢——抢，你会反抗；不是骗——骗，你事后会发现。偷的特点是，它发生在默认选项里。你什么都不用做，它就完成了。

法币偷走你的过去。储蓄是你已经付出的时间的凝结：你用若干年的清晨和深夜换来一个数字，这个数字代表的是时间，不是纸。而法币体系唯一真正的自由，就是增发。每一次增发都不改变你账户里的数字，只改变这个数字能换回多少时间。通胀不是经济现象，它是针对储蓄者的时间盗窃——用最温和的方式，拿走你最不可再生的东西。

算法偷走你的现在。注意力是时间在当下的形态：你此刻在看什么，你此刻就活在什么里面。推荐系统的生意，是把这段时间切碎、拍卖、再切碎。每一次刷新都是一次竞价，竞的是你接下来十五秒的生命。你不是竞价的一方，你是标的。

AI 偷走你的未来。专业知识曾经是你对未来时间的定价权：你学了十年，所以你的一小时值这么多。当认知可以被无限复制，每一次复制的边际成本趋近于零，定价权就不再在你手里。这不意味着你会失业。这意味着你的未来时间会被别人定价，而且定得很便宜。

三者有一个共同点：它们都不需要你同意。你没有签过允许央行稀释你储蓄的协议，没有签过允许算法拍卖你注意力的协议，更没有签过允许机器用你的产出训练自己、再回头给你定价的协议。留在里面不需要任何动作，退出每一种都需要主动动作。这就是偷。

三、梯度

宇宙的铁律是熵增。这句话被说得太多，以至于多数人得出了错误的结论：既然秩序注定瓦解，一切努力都是悲壮的抗争。

生命从来不是这样活下来的。生命是开放系统：它不对抗熵增，它消耗一个能量梯度，把熵排到外面去，换取内部的秩序。太阳和地球之间的温差养活了所有生物；一碗饭和一个饥饿的身体之间的梯度养活了你。熵增是宇宙的税，秩序是你用梯度买来的东西。没有人能免税，但每个人可以决定自己的梯度归谁。

所以时间领主的问题从来不是“如何抗争”，而是两个更具体的问题：你的梯度是什么？它是否正在被别人收走？

你只有三个梯度。身体，是能量转化为时间的地方，它的效率决定你能用多少小时。注意力，是时间转化为秩序的地方，它的完整度决定你的小时能产出什么。部落，是秩序得以复利的地方：一个人的秩序会随他死去，一群人的秩序可以传下去。三重盗窃偷的，恰好就是这三个梯度——法币抽走你用身体换来的储备，算法切碎你的注意力，AI 让你的专业不再足以聚起一个部落。

守住梯度，不是自我提升的鸡汤，是热力学。

四、分配

旧的分配机制是劳动：你出时间，换回份额。这条链运行了一万年。它不公平，但它清楚——一个人占有多少，大体上和他投入了多少时间有关。工资、工时、退休金、劳动法，都是在这条链上做修补。

当机器出时间，这条链就断了。一台机器可以在一秒内供给一万小时的认知劳动，而这一万小时不属于任何人的生命。AI 之后，生产不再稀缺，稀缺的是分配。机器放大一切产出，却没有任何机制规定放大的部分归谁——默认归谁，就是归拥有机器的人。

于是一个文明可以同时拥有无限的产出和无处安放的人。这不是技术问题，技术已经解决了生产。这是账本问题：我们的账本只会记“谁付出了劳动”，不会记“谁存在”。当劳动不再是分配的依据，账本就必须换一个依据。

这里有一条我认为是硬约束的原则：捕获价值的机制，不能同时是分配价值的机制。拥有机器的人决定机器的产出归谁，这不叫分配，这叫圈地。捕获与分配必须分离。这是我衡量任何分配方案的尺子。

五、协议

换一种货币，不够。比特币改变了货币的发行权，模因币取消了发行的门槛，稳定币干脆把法币本身搬上了链。它们都在某个维度上成功了，但没有一个逃出法币：只要一种货币的涨跌仍由资金决定，它的价值就仍以法币计价，它的参与者就仍在玩赌场的游戏。这场革命不彻底，不是因为技术，是因为它们只换了账本上的单位，没有换账本记的东西——记的仍然是“谁持有多少”，而不是“谁存在”。

已经有人把另一种思路写成了协议，从 2023 年起在几条链上做实验。我把它当作众多可能的分配方案之一来看。它的设计分三层。

第一层，时间存在证明。把存在本身记录为不可稀释的凭证。存在不是身份，不是资产，不是你拥有什么或做了什么，是你在时间中出现过这个事实。这是链下世界最硬的事实，也是唯一一个机器无法替你生成的事实。把它记上账，分配就有了一个不依赖劳动的依据。这一层可以有很多种实现，这个协议只是其中一种。

第二层，注意力与创造力的市场。在存在凭证之上，让人的注意力和创造力被定价——由出让的人定价，而不是由平台定价。这一层把算法偷走的现在，变回你可以自己出售的现在。

第三层，AI 生产税。让每一份被机器放大的产出，向人的时间纳税。不是向政府纳税，是向存在凭证的持有者纳税——向所有在时间中出现过的人。机器用一万小时认知劳动创造的价值，回流到那些被机器定价的人手里。

它暴露了这类协议都绕不开的问题：存在证明如何抵御伪造的存在，税率由谁决定，市场在没有流动性的时候如何定价。这些问题还没有全部解决。它也证明了可能：一个不依赖劳动的分配依据，可以被写成代码，可以被几百个陌生人使用，可以在几条链上活过近四年。

它不是唯一的解，也未必是最好的解，但它是一个可以被验证的解。我更希望看到你提出自己的分配协议，而不是相信这一个。

六、终局

终局不是逃离，是延续。

主权个人的叙事通常止步于逃离：换一个国家，换一种货币，换一个身份。这是把时间主权理解成了迁徙自由。迁徙解决不了时间的问题——你把自己搬到哪里，时间都以同样的速度流走。

肉体的终点由生物学决定，影响力的终点由账本决定。一个人死后，他的思想、决策、作品是否继续起作用，取决于它们被记在什么地方。记在平台上的，随平台消失；记在硬盘上的，随硬盘损坏；记在后人记忆里的，三代以内归零。互联网承诺了记忆，只留下了遗忘。

把它们刻进永不关闭的档案——一个靠数学和经济激励、而不是靠任何机构的善意来保存的地方。让它们在你之后继续被读取、被执行、被继承：你的判断可以被训练成一个继续替你判断的代理，你的存在凭证可以指定由谁继承，你的账本可以在你不在场的时候继续结算。

如果时间是货币，死亡就是最终的破产。而代码第一次给了我们重组的机会——不是不死，是不被清零。

七、选择

你无法选择时间是否流走。这一点，物理已经替你决定了。

你只能选择它记在谁的账上。记在央行的账上，它是通胀；记在平台的账上，它是流量；记在机器拥有者的账上，它是算力的折旧。记在你自己的账上，它才是你的一生。

这份宣言不要求你相信任何东西。它只要求你去看一眼那笔账。

从今天起，记在你自己的账上。

三重盗窃

时间盗窃系列 · 一 · 展开《时间主权宣言》第二节

> 三套系统，先后学会了偷时间。三者有一个共同点：它们都不需要你同意。 ——《时间主权宣言》

为什么是"偷"

这个字太重，所以先把它说清楚。

抢，是显性的。有人当面拿走你的东西，你知道发生了什么，你可以反抗，法律站在你这边。骗，是延迟的。你当时同意了，事后发现同意的前提是假的，你可以追责。偷，和这两者都不一样：它发生在你不在场的时候，发生在默认选项里。你什么都没做，它就完成了；你什么都没察觉，因为它从一开始就被设计成不需要你察觉。

我用"偷"而不用更客气的词，是因为所有客气的词都把责任推回给你。"通胀是经济周期"——所以你该学会理财。"刷手机是自制力差"——所以你该自律。"被 AI 替代是没有终身学习"——所以你该再学一门。三句话的结构一模一样：系统拿走了你的东西，然后告诉你这是你的问题。

"偷"这个字把责任放回系统。它的定义很具体：未经同意的转移。下面三节，每一节讲一套系统怎样完成这种转移，拿走的是你时间的哪一部分，以及为什么你从来没有被问过。

一、法币偷走你的过去

先弄清楚储蓄是什么。

储蓄不是钱。储蓄是你已经花掉的时间的余额。你在某一年的某几百个清晨和深夜里做了别人需要的事，换来一个数字，这个数字代表的是那几百个清晨和深夜——它们已经过去了，不可能再来一次。所以储蓄是过去时。它是你生命里已经支出、尚未兑现的那一部分。

1971 年 8 月 15 日，美国关闭了美元与黄金的兑换窗口。从那一天起，世界上主要货币的数量不再受任何物理约束，只受政策约束。这件事的意义不在于黄金本身——黄金只是一个约束的载体——而在于约束消失之后，货币体系获得了一种此前没有的自由：增发的自由。

增发是怎么偷的？它不动你账户里的数字。你的一百块还是一百块。它改变的是这一百块能换回多少小时的别人的劳动。按美国劳工统计局的消费价格指数，1971 年的 100 美元相当于 2026 年的约 832 美元——反过来说，一个在 1971 年存下 100 美元现金、一直持有到今天的人，手里的

购买力只剩下原来的 12%。年均 3.9%，听上去温和，五十五年后剩下八分之一。任何法币都一样，差别只是速度。

经济学有一个词叫铸币税：发行者用几乎为零的成本创造新货币，再用它购买真实的商品和劳动。这笔税的税基是所有持有该货币的人，税率是增发的幅度，征收方式是购买力的稀释。它从来不出现在任何一张税单上。

这笔税还有一个特点：它是累退的。新货币不是均匀地落到每个人头上，它从发行点出发，先经过银行、政府、大资产的持有者，最后才到工资和储蓄。先拿到的人用旧价格买东西，后拿到的人用新价格买东西——这就是所谓的坎蒂隆效应。资产持有者在通胀中得到补偿，因为资产随货币一起涨；靠工资和现金储蓄生活的人得不到，因为工资总是最后一个涨的。所以铸币税不仅是一笔没有税单的税，还是一笔主要由耐心的人和没有资产的人缴纳的税。

这里有一个更隐蔽的层次。欧文·费雪在 1930 年把利息解释为"等待的报酬"：你推迟消费，把资源让渡给别人先用，利息是对你的耐心的补偿。而当真实利率为负——增发的速度超过利息——等待不再被补偿，而是被惩罚。一个体系如果持续地惩罚耐心、奖励即时消费和负债，它偷走的就不只是储蓄的购买力，还有储蓄这件事本身的意义。它在系统性地教你不要为未来留东西。

反方的论证值得认真对待：温和的通胀是必要的，通缩更糟——物价下跌让人推迟消费，企业收入下降，裁员，再下跌，螺旋。这个论证在宏观上有它的道理，我不打算在这里反驳它。但它回答的是另一个问题。它回答的是"体系需要什么"，不是"谁来付"。即使温和通胀对体系整体是最优的，这个最优也是靠一笔没有人同意、没有人看见、主要由储蓄者缴纳的税来实现的。一个必要的税也仍然是税；一个没有税单的税仍然需要有人问一句：谁同意了？

没有人问过你。储蓄者的"同意"是通过"持有这种货币"被默认推定的，而你出生在一个法币国家，持有它不是选择，是起点。你唯一的退出方式是把它换成别的东西——换什么、什么时候换、换错了怎么办，成本和风险全部由你承担。留在里面不需要任何动作；离开需要知识、判断和运气。这就是默认选项的力量。

顺带一提，1971 年还有另一本书出版：乔治斯库-罗根的《熵定律与经济过程》。他说经济过程本质上是一个熵过程——低熵的资源进去，高熵的废物出来，不可逆。这和本系列的底层框架相通，留到后面的篇章展开。但两件事发生在同一年，不是巧合，是那个年代的人已经隐约感到：约束一旦解除，有些东西就只会单向流走。

二、算法偷走你的现在

同样是 1971 年，赫伯特·西蒙写下一句后来被引用了无数次的话：信息的丰富造成注意力的贫乏。他的本意是设计——既然注意力是稀缺品，信息系统就应该被设计成节约注意力的工具。五十年后，信息系统的设计目标变成了相反的东西：尽可能多地消耗它。

注意力是什么？它是时间的当下形态。你此刻在看什么，你此刻就活在什么里面。过去已经支出，未来还没到，你真正拥有的只有正在流逝的这一秒，而这一秒的内容就是你的注意力所在。所以偷注意力，偷的不是时间的某个部分，是你活着的唯一时态。

它是怎么被偷的？用拍卖。

互联网广告的核心机制是一种实时拍卖。2007 年，埃德尔曼、奥斯特罗夫斯基和施瓦茨在《美国经济评论》上描述了谷歌和雅虎使用的"广义第二价格拍卖"：每一个广告位在每一次展示时被拍卖一次，出价最高者得到它，付第二高的价格。后来的机制更复杂，但骨架没变。你每刷新一次页面，就有一场拍卖在几十毫秒内完成，标的是"接下来几秒钟内这个人的注意力"。

注意这场拍卖的参与方：买方是广告主，卖方是平台，拍卖师是算法。你是什么？你是标的。标的不参与定价，不分享收入，甚至不知道拍卖发生过。

推荐系统在这个结构里的角色就清楚了。它的目标函数不是"为你找到最好的内容"，而是"让你停留最长的时间"，因为停留时长决定拍卖的场次。它不是在为你找内容，它是在为广告找你。算法之所以显得懂你，是因为懂你是让你停留的最短路径。

所以产品的每一个细节都不是中性的。无限下滑取消了"翻到底"这个天然的停止信号；自动播放取消了"按下一个"这个决定；红点和震动把一个本来由你发起的动作——"我去看看有没有消息"——变成一个由它发起的动作。这些设计没有一个是为了帮你决定什么时候停，它们全部是为了取消"停"这个选项的存在。一个人被拍卖的时长，等于他找不到停止键的时长。

代价是可以测量的。加州大学欧文分校的格洛丽亚·马克和同事在 2008 年的研究里让人们在工作中被打断，然后测量他们回到原任务的时间：平均大约二十三分钟。更值得注意的是她的第二个发现：人们会通过加快速度来补偿被打断的时间，代价是更高的压力和挫败感。也就是说，碎片化不只是偷走被打断的那几分钟，它偷走的是之后的二十三分钟，以及你为了追回这二十三分钟付出的额外消耗。

反方会说：服务是免费的，这是交换，不是偷。这个说法有两处不成立。第一，价格不是零——你付的是你唯一不可再生的东西，只是账单上不写这一项。第二，交换需要双方定价，而这里只有一方定价：平台决定你的一小时注意力值多少广告，你既不知道这个数，也无权议价。一场只

有一方知道价格的交换，在任何市场里都不叫交换。

没有人问过你。你"同意"过的是一份没有人读的服务条款，而同意的默认结果是：你的注意力被拍卖，拍卖的收入归平台，你得到的是免费的内容——也就是说，你以为你在消费，实际上你在被消费。退出需要卸载、需要自制、需要承受社交上的缺席；留下只需要什么都不做。

三、AI 偷走你的未来

专业知识是什么？它是你对未来时间的定价权。

一个人用十年学一门专业，不是为了那十年本身，是为了之后的三十年里，他的一小时能卖出一个比别人高的价格。这个价格是他对自己未来时间的定价，而定价的依据是：这件事别人做不了，或者做不了这么好。专业知识是一份关于未来的合同，用过去的十年买下来的。

现在看这份合同的另一方发生了什么。

2024 年 11 月，a16z 的一篇分析统计了大语言模型的推理价格：同等能力的模型，每百万词元的推理成本在三年里从 60 美元降到了 0.06 美元，千倍；换算成速率，大约每年下降十倍。这个速度超过了个人电脑时代算力成本的下降速度。具体的数字每个月都在变，方向不会变：认知劳动的边际成本正在趋近于零。

高盛在 2023 年 3 月的报告里估计，生成式 AI 可能让相当于三亿个全职岗位暴露在自动化之下，受影响最深的是办公室和行政类工作——也就是那些靠专业知识定价的工作。

但失业不是这件事的核心。多数人不会在某一天被通知"你被替代了"。发生的事情更安静：你的一小时会被重新定价。新的定价依据不再是"别人做不了"，而是"机器做这件事要多少钱"。你用过去十年买下的那份未来合同，被单方面重新估值了，而估值的人不是你。

反方的论证是历史的：每一次技术革命都消灭了一批工作，也创造了一批新的，蒸汽机、电力、计算机都是如此，这一次不会例外。我希望它是对的，但有一个结构上的差别不能略过。以前的每一次替代，替代的都是人类卖得便宜的东西——肌肉、重复、记忆——然后把人推向卖得更贵的东西：判断、创造、专业。每一次替代都抬高了认知的价格。这一次替代的是认知本身。"推向更贵的东西"这个机制，需要有一个更贵的东西存在；这一次，我们还没有看到那一级台阶在哪里。也许它会出现。但在它出现之前，被重新定价的人没有地方可去，而重新定价已经发生了。

这是对"未来"的偷。法币偷的是你已经支出的时间，算法偷的是你正在支出的时间，AI 偷的是你还没支出、但已经定好价的时间。

还有更深的一层。这些模型是用人类的产出训练的——文章、代码、对话、作品，其中包括你的。你的产出先被拿去训练，训练出来的东西再回头给你定价。两次转移，两次都没有问过你。第一次拿走的是你过去的产出，第二次拿走的是你未来的定价权。而所谓"选择退出"，和前两种盗窃一样，是一个需要你主动发现、主动申请、主动证明的选项——默认是进。

这不是一个道德指控，是一个结构描述：在这条链上的每一个环节，你的时间都不是一个变量。

四、它们都不需要你同意

把三件事放在一起看，结构是一样的。

每一种盗窃都是一个默认选项。留在里面不需要任何动作：你出生在法币里，你打开手机就在算法里，你的产出发布出去就在训练数据里。退出每一种都需要主动动作，而且动作的成本——知识、时间、风险、社交代价——全部由你承担。

每一种盗窃都把责任转回给你。通胀教你理财，算法教你自律，AI 教你终身学习。这三句话都对，也都不是重点。重点是：在三套系统各自的目标函数里，你的时间根本不是一项。央行的目标是物价和就业，平台的目标是停留时长，模型公司的目标是能力和成本。没有人想偷你的时间。它只是不在任何人的账本上。

那么"同意"应该长什么样？三个检验，每一个都很朴素：

第一，看得见。转移发生的时候，你能看到它——金额、时点、去向。铸币税没有税单，拍卖不通知标的，训练不通知作者。三个都不及格。

第二，能说不，而且说不的成本为零。一个需要你花钱、花时间、冒风险才能拒绝的选项，不是选项，是收费站。换货币、戒手机、申请退出训练——三个都要收费。

第三，你的时间在账本上是一个变量。只有当一笔转移被记录在一个以你的时间为单位的账本上，"同意"才有可能被定义——否则你连要同意什么都说不清楚。三套系统的账本上，这一项都不存在。

三重盗窃不是三个坏人干的三件坏事，它们是同一个缺口的三种表现：我们的账本只记"谁持有多少"和"谁付出了劳动"，不记"谁存在"、"谁的时间"。当货币可以无限增发，当注意力可以被拍卖而不付给本人，当认知可以被复制而不付给原作者，这三件事能够发生的共同前提是——时间在账本上没有名字。

所以问题不是"怎么反抗"。反抗一个不把你当变量的系统，就像向一台机器抗议：它听不见，不是因为它恶意，是因为它没有那个输入端口。

问题是：账本该记什么。

这是第二篇的主题。当机器开始出时间，劳动就不再是分配的依据——而旧账本上，除了劳动，什么都没有。

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法币偷走你的过去，算法偷走你的现在，AI 偷走你的未来；时间主权，就是把三者夺回来。

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分配

时间盗窃系列 · 二 · 展开《时间主权宣言》第四节

> 旧的分配机制是劳动：你出时间，换回份额。当机器出时间，这条链就断了。 ——《时间主权宣言》

一万年的那条链

上一篇的结论是：三重盗窃能够发生，是因为时间在账本上没有名字。这一篇讲账本本来记什么，为什么它快要记不下去了，以及换一种货币为什么救不了它。

人类历史上只有过一种大规模的分配机制：劳动。你出时间，换回份额。狩猎采集的时代，分享是规范，但分享的依据仍然是谁出了力；农业把这条链固定下来——土地加劳动，一万年；工业把它提纯成最赤裸的形态——计时工资，你卖的东西不是产品，是小时。这条链有一个了不起的美德：它清楚。它不公平，资本、土地、出身都在它之外另开账户，但一个人占有多少，大体上和他投入了多少时间有关。过去两百年所有关于公平的斗争——八小时工作制、最低工资、养老金、工会、劳动法——没有一项试图换掉这条链，全部是在链上做修补。

1930 年，凯恩斯写了一篇短文《我们孙辈的经济可能性》。他预言一百年后，技术进步会把生产问题解决掉，人类每周只需要工作十五个小时，剩下的难题是如何打发闲暇。他看到了一半。生产问题确实在被解决——从他写作到现在，人均产出翻了好几倍。但每周十五小时没有到来，原因不在技术，在账本：份额系在劳动上，不工作就没有份额，所以没有人敢少工作。凯恩斯假定分配会自动跟上生产，这个假定错了，因为他没有看到链本身需要换。

链的磨损在 AI 之前就开始了。卡拉巴布尼斯和奈曼在 2014 年的研究里统计了全球的劳动份额——国民收入中归劳动者的比例——发现它从 1980 年代初以来下降了大约五个百分点，而且是全球性的，不限于某个国家或行业。另一组数字更直观：美国经济政策研究所长期追踪的"生产率与报酬的缺口"显示，1979 年以来美国的劳动生产率上升了约八成，而普通工人的时薪报酬只上升了约三成——在那之前的三十年，两条线是贴着走的。产出和份额在 1970 年代末分了手，而那正是第一篇讲的法币约束解除之后的几年。资本替代劳动，已经悄悄进行了四十年。AI 不是这个趋势的起点，是它的加速器。

当机器出时间

一台机器可以在一秒内供给一万小时的认知劳动，而这一万小时不属于任何人的生命。这句话里有一个此前从未出现过的东西：没有生命的时间。

劳动链之所以成立，是因为时间是从人的生命里出来的——每一小时劳动背后都有一个人少活了这一小时，份额是对这个消耗的补偿。当时间可以不从生命里出来，补偿的对象就消失了。机器出时间，谁该拿份额？按劳动链的逻辑，是机器的所有者，因为是他"投入"了机器。但他投入的是资本，不是时间；而资本链从来不是一条分配链，是一条集中链——资本的回报归资本，这是它的定义。

科里内克和斯蒂格利茨在 2017 年的论文里把这个问题说得很清楚：AI 对收入分配的影响不由技术决定，由制度决定；在现有制度下，最可能的结果是劳动份额进一步下降，工资被压向机器的成本。他们提出了各种再分配的工具，但所有的工具都是在旧账本上做转移支付——先让链断掉，再用税收把一部分份额搬回去。

赫拉利把被这条链甩出去的人叫作"无用阶级"。这个词很准确，也很残酷，但它准确的方式值得注意：它是从劳动链的视角看人。一个人在劳动账本上没有条目，他就是"无用"的。换一个账本，同一个人不是无用的，他是存在的。"无用"不是对人的判断，是对账本的判断——它说明这本账只认一种输入。

于是出现一个此前从未有过的组合：无限的产出，和无处安放的人。内容可以零边际成本无限生成，常规的认知服务——法律、会计、编程、翻译的大部分——正在变成一种公用事业。生产不稀缺了。稀缺的是分配：这些产出归谁，依据什么。而我们的账本只会记两件事——谁持有多少（资本），谁付出了劳动（劳动）。资本在集中，劳动在贬值，账本上没有第三栏。

这不是技术问题。技术已经解决了生产，它解决得太好了。这是账本问题：当劳动不再是分配的依据，账本必须换一个依据，而换依据的事，技术不会自动做。

有人会说：那就让每个人都拥有机器。员工持股、全民基本资本、主权财富基金——挪威把石油收入放进一个全民共有的基金，这是现成的例子。这个方向比转移支付好，因为它触及了捕获的源头。但注意它换的是什么：它让更多人进入"谁持有多少"那一栏，分配依据仍然是持有。持有是会集中的——这是资本的性质，不是道德缺陷——所以按持有分配，长期会回到起点，除非持有本身被绑定在某个不会集中的东西上。什么东西不会集中？一个人只能存在一次。这就是为什么分配依据必须换到"存在"上，而不是换到"更多人持有"上。

账本也不是完全没有记过"存在"。人口普查记存在，国籍记存在，近十年各地的基本收入试验——芬兰 2017 到 2018 年的两千人试验、美国斯托克顿 2019 年起的一百二十五人试验、"直接捐赠"在肯尼亚的长期试验——都是按存在而不是按劳动发放份额的尝试，它们的结果大体一致：收入稳定了，就业没有减少，心理健康改善。它们证明了按存在分配在人的层面是可行的。它们的限制在另一边：钱从哪里来。每一个试验都由一个捕获者——政府或基金会——出资，所以它们是转移，不是分配机制；出资者一停，份额就停。这就引出了下面的问题：如果分配依据要换到存在，那么分配的钱必须从捕获的源头直接来，而不是经过捕获者的手。

换一种货币，为什么不够

读到这里，很多人会想到一个现成的答案：换一种货币。过去十几年里，三种尝试先后出现，每一种都值得认真对待，每一种都不够。

比特币改变了货币的发行权。2008 年中本聪的白皮书给出了一个固定供应、无人能增发的货币，这是对第一重盗窃最直接的回应：储蓄不再被稀释。它做到了它承诺的事。但注意它改变了什么、没有改变什么——它换掉了账本上的单位，账本记的东西没有变，仍然是"谁持有多少"。比特币的分配依据是算力和进入的早晚，这是资本链和时机的混合，不是一个新的依据。它是更好的储蓄技术，不是另一种分配。

稳定币干脆把法币本身搬上了链。到 2026 年，这个市场有三千多亿美元的规模，它证明了链上结算的效率，也证明了一件更有意思的事：当人们可以自由选择链上的单位时，多数人选了美元。稳定币是效率的胜利，是主权的放弃——它连单位都没换。

模因币取消了发行的门槛。任何人花几分钟就可以发行一种货币，这把"发行权"民主化到了荒谬的程度：几百万个微型央行同时存在。这场实验有一个意外的教学价值——它证明了发行自由不等于分配正义。人人都能发币，分配的规则却和一百年前一模一样：谁先进、谁先拿、谁走得快谁赢。

三者有一个共同点：只要一种货币的涨跌仍由资金决定，它的价值就仍以法币计价，它的参与者就仍在玩赌场的游戏——零和、波动、靠注意力驱动。这场革命不彻底，不是因为技术不够，是因为它们只换了账本上的单位，没有换账本记的东西。

比特币的信徒会反驳：储蓄技术不需要解决分配，它只需要做好一件事。这个反驳是对的，而且正是本系列的论点——储蓄技术解决的是第一重盗窃，它不解决第二重和第三重，也不解决 AI 之后的分配。它是必要的，不是充分的。一个只能看见持有者的账本，看见的不是人，是余额；在这样的账本上，不持有就等于不存在。

捕获与分配必须分离

现在可以说出这个系列最核心的一条原则。先定义两个词。

捕获，是把产出变成可支配的价值——谁拿到了收入。分配，是决定价值归谁。在劳动链里，这两件事是由同一个动作完成的：工资既是捕获（企业付出），也是分配（劳动者得到），市场把两者合并执行，所以没有人需要区分它们。

机器时代把它们拆开了。捕获集中到机器的所有者手里——这是资本的性质，无可厚非。问题在于分配：如果分配也由机器所有者决定，那么机器创造的一切产出，默认归机器所有者，其余的人得到的是他愿意给的部分。这不叫分配，这叫圈地。十六到十八世纪的英格兰，公地被圈成私产，原本靠公地生活的人变成无地的劳动者，波兰尼把这称为劳动被商品化的起点。今天被圈的公地是认知：模型用全人类的产出训练，产出归训练模型的人。

所以这条原则是：捕获价值的机制，不能同时是分配价值的机制。

这不是一个新直觉，它是人类最成熟的几项制度设计背后的共同直觉。孟德斯鸠的三权分立：立法者不能同时是执法者。伯利和米恩斯 1932 年描述的所有权与控制权分离：出资的人和经营的人不能是同一个意志。央行独立：花钱的部门不能同时是印钱的部门。审计独立：记账的人不能审自己的账。每一项都在说同一句话——裁判不能同时是运动员。

这条原则有一个经常被忽略的推论：国家也是捕获者。税收就是捕获。所以"交给国家来分配"并不自动满足分离——它只是把分配权从一个捕获者交给另一个捕获者。真正的分离要求分配机制对所有捕获者都独立，无论是企业还是国家。这就是为什么接下来要讨论的是协议而不是政策：政策是捕获者的工具，协议可以不是。

把这一篇和上一篇合起来，就得到一把尺子。任何一个声称解决"AI 之后的分配"的方案，可以用四条来量：

第一，它的分配依据是否不依赖劳动。如果还是按劳动分配，它只是在修补一条已经断掉的链。

第二，它的捕获与分配是否分离。如果分配者同时是捕获者——企业或国家——它是圈地的某种改良。

第三，它是否通过"同意"的三个检验：转移看得见，说不的成本为零，时间在账本上是一个变量。

第四，它是否可以被验证。一个无法验证的分配方案，是一个承诺，不是一个机制。

下一篇要讲的是：已经有人试着把这四条写成了协议。我会用这把尺子去量它，而不是替它说话。

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法币偷走你的过去，算法偷走你的现在，AI 偷走你的未来；时间主权，就是把三者夺回来。

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一种协议

时间盗窃系列 · 三 · 展开《时间主权宣言》第五节

> 已经有人把另一种思路写成了协议，从 2023 年起在几条链上做实验。我把它当作众多可能的分配方案之一来看。它的设计分三层。 ——《时间主权宣言》

先说立场

上一篇结尾给出了一把尺子：分配依据不依赖劳动；捕获与分配分离；通过"同意"的三个检验；可以被验证。这一篇用它来量一个具体的东西。

我不是这个协议的代言人，也不打算替它说话。我把它当作众多可能的分配方案之一，之所以选它来量，是因为它是我所知道的、把"存在"而不是"劳动"或"持有"作为分配依据的少数几个已经在链上运行的尝试之一。量完之后，你会看到它哪里站得住，哪里还只是一个声明。我更希望这一篇的结果是：你去设计一个更好的。

它的设计分三层。逐层来看。

第一层：时间存在证明

要让"谁存在"进入账本，必须先回答两个问题：怎样证明一个存在是真的，怎样不把"存在"变成"身份"。

第一个问题在密码学里有名字，叫女巫攻击：一个人伪装成一千个人。博尔赫等人在 2017 年提出"人格证明"这个概念时，目标就是在没有中心化身份机构的前提下，做到一人一份。此后出现了三条主要路线。生物特征路线：用虹膜或其他生理特征保证唯一性，代价是一个中心化的采集方，和一个一旦泄露就无法更换的密钥。社会图谱路线：让已经被确认的人互相担保，代价是串通——一群人可以互相担保出不存在的人。活性测试路线：要求所有人在同一时刻完成某种机器难以通过的任务，代价是随着机器变强，测试一个一个失效。三条路线没有一条完全解决问题；这是全世界范围内都没有解决的问题，不是某个协议的失败。

第二个问题——存在不等于身份——有一个精确的工具。1985 年，戈德瓦瑟、米卡利和拉科夫提出零知识证明：你可以向别人证明一个陈述为真，而不透露这个陈述之外的任何信息。应用到这里就是：你可以证明"我是一个真实存在的人，而且不是已经登记过的任何一个"，而不透露你是谁。存在被记录，身份被保护。这不是理论上的可能性，是已经在用的技术。

这一层真正的设计点，在"时间"两个字上。多数人格证明方案把"存在"当作一个布尔值：是人，或者不是人，然后一人一票或一人一份。这个协议的凭证是按时间累积的——每一个周期在场一次，账本上多一份。于是"存在"从一个是非题变成了一条时间序列：账本记的不是"这是一个人"，而是"这个人在这些时间里在场"。时间成了单位。这和第一篇的诉求是对应的：时间在账本上终于有了名字。

它对女巫问题的处理，据我理解，不是宣称解决，而是一个工程取舍：先承认存在证明不可能完美，再用后两层的经济结构让伪造存在无利可图——只要伪造一个存在的成本高于这个存在能分到的份额，伪造就是亏本的。这是一个合理的取舍，不是一个数学证明。它成不成立，取决于后两层的参数，而参数是会变的。这是第一个必须盯住的地方。

还要把它和两个容易混淆的东西分开。它不是基本收入。基本收入是一笔转移：由一个捕获者——政府、基金会、或者一家用虹膜换代币的公司——出钱，按人头发放，出资者停，份额就停。存在凭证不是一笔钱，是账本上的一栏；钱从哪里来是第三层的事，第一层只负责让"谁存在"被记下来。它也不是一人一票。投票是把存在当作一个布尔值来用；按时间累积的凭证把存在当作一个数量来记——在场了多久，就记多少。前者回答"你是不是人"，后者回答"你的时间在哪里"。这两个区别决定了它和过去十年所有"人格证明"项目的不同：那些项目在证明人，这个协议在记录时间。

第二层：注意力与创造力的市场

第一篇描述过一场拍卖：买方是广告主，卖方是平台，你是标的。第二层仍然是一个市场，参与方换了：卖方是人。

这个想法也有谱系。拉尼尔在 2013 年的《谁拥有未来》里提出，每一份被使用的个人数据都应该给原主人一笔微支付。2018 年，阿列塔-伊巴拉、戈夫、希门尼斯-埃尔南德斯、拉尼尔和韦尔在《美国经济评论》论文集上把它表述为一个更锋利的问题："数据应该被当作劳动吗？"——如果机器学习的原料是人的产出，那么提供原料的人就是劳动者，应该被定价。2017 年，Brave 浏览器的注意力代币做了一次局部实现：用户看广告，分到一部分收入。它证明了可行，也暴露了局限：它仍在广告框架里，仍由平台撮合，用户得到的是平台决定的分成。

这一层的设计点有三个。定价权在出让者手里，不在平台手里——你决定你的一小时注意力值多少，买方接受或者不接受。只有持有存在凭证的人才能出让——存在是入场券，这同时也是第一层对第二层的保护：机器刷不出注意力，因为机器没有凭证。创造力和注意力进同一个市场——作品和在场是同一种东西的两种形态，都是时间。

它的问题也很清楚：市场需要买方。一个没有买方的市场只是一个价签。谁来买人的注意力和创造力？如果答案是"需要人类数据来训练的模型公司"，那么这一层和第三层就是连通的——税和市场是同一笔钱的两种流向。如果没有买方，这一层就只是一个设计。这是第二个必须盯住的地方。

第三层：AI 生产税

"对机器征税"这个想法不新。盖茨在 2017 年提过机器人税：替代工人的机器人应该像工人一样纳税。科里内克提过"AI 红利"。加州在 2019 年讨论过"数据红利"。所有这些的直觉都一样：机器放大的产出里有一部分是公共的——训练数据是全人类的产出，没有哪一家公司创造了它——所以放大的部分不应全部归放大者。

有一个先例值得看，因为它已经运行了四十多年：阿拉斯加永久基金。1982 年起，阿拉斯加把石油收入的一部分放进基金，每年向每一个居民发放分红——2025 年是每人一千美元。它的逻辑和这里完全一致：石油是谁都没有创造的价值，所以按存在分配，而不是按劳动。阿拉斯加的分红依据就是"你在这里住了一年"——时间存在证明的一个行政版本。

这个协议的第三层和以上所有方案有两处差别。第一，税不交给政府，交给存在凭证的持有者。原因在上一篇：国家也是捕获者，税收经过国家，就是经过了一个捕获者的手，分离就不成立了。第二，税基是"被机器放大的产出"，不是机器本身——机器人税的定义困难（什么算一个机器人？）被绕开了，但换来另一个困难：什么算"被机器放大"？

这一层没有解决的问题最多：税率由谁决定，这是治理问题；税基怎么测量，这是技术问题；链上的税怎么触及链下的产出，这是管辖问题。诚实地说，第三层目前更像一个声明而不是一个机制。它说清楚了钱该从哪里来、到哪里去，还没有说清楚怎么收。这是第三个必须盯住的地方。

近四年的实验

这个协议从 2023 年起在几条链上运行，被几百个陌生人使用。近四年能证明什么，不能证明什么，要分开说。

它证明了：一个不依赖劳动的分配依据可以被写成代码；可以被素不相识的人使用；可以在几条链上活过近四年而不依赖任何机构。这三件事在 2023 年之前都只是猜想。

它没有证明：规模——几百人不是几亿人；女巫抵抗——没有人认真攻击过它，不等于它抵抗得住；流动性——市场在没有买方的时候如何定价，近四年没有给出答案；法律地位——一个向存在凭证持有者"纳税"的协议，在任何一个法域里都还没有名字。

还有一件事必须说：设计与部署之间有距离。三层是设计，链上跑的是实现，实现未必覆盖全部设计。读者应当按链上的事实判断，而不是按文字。这一条适用于一切协议，包括这一个。

用上一篇的尺子量一遍：

分配依据不依赖劳动——设计上成立。它的依据是在时间中的存在。

捕获与分配分离——设计上成立。税不经过捕获者的手，直接到存在凭证的持有者。但"设计上"三个字要重读：如果税率由少数人决定，分配权就回到了某个捕获者手里。

同意的三个检验——看得见，成立，链上的转移是公开的；说不的成本为零，部分成立，退出容易，但进入需要先证明存在，而证明是有成本的；时间是账本上的变量，成立，这是它和其他方案最根本的区别。

可以被验证——成立。链上的一切都可以被任何人查看。

四条里，它在设计上全部通过，在运行上只通过了"可验证"一条，其余三条都停在"设计上成立"。这就是它现在的位置：一个可以被验证的解，不是一个被证明的解。

什么会杀死它

既然要用第三方的眼光看，就把它可能失败的方式列出来，按我判断的可能性排序。

女巫经济翻转。只要伪造一个存在的成本降到它能分到的份额以下，伪造就从亏本变成盈利，账本会在几周内被假存在淹没。机器正在让"伪造一个像人的在场"越来越便宜，这条线每年都在逼近。

税率被俘获。第三层的税率由谁定，是整个设计最薄的地方。如果由少数持有者决定，他们会把税率调向对自己有利的方向；如果由多数决定，多数会投票给自己加税——两种结果都是分配权回到某个捕获者手里，分离原则失效。目前没有已知的治理机制能同时避免这两种结果。

没有买方。第二层的市场如果长期只有卖方，凭证就只是一个记录，没有兑现的途径。记录本身有价值——它是时间在账本上的名字——但一个不能兑现的名字，留不住使用者。

法律定性。一个向凭证持有者"纳税"的链上协议，在多数法域里会被按证券、按支付工具、或按彩票来定性，三种定性都会让它在该法域内不能合法运行。这不是协议能自己解决的，它取决于协议之外的人。

设计与部署的距离。三层里有多少已经在链上，多少还在文档里，这件事只能查链，不能听人说。

把这五条列出来，不是为了唱衰。一个能被列出失败方式的方案，比一个不能被列出的方案更可信——前者是机制，后者是信仰。

怎样查一个协议

顺便给一个办法，适用于任何声称解决分配的链上协议，不只是这一个。打开区块浏览器，查五件事：合约的创建时间，它决定"运行了多久"是真是假；持有人数和转账数，它决定"被多少人使用"是真是假；源码是否经过验证，它决定你能不能看到它实际做的事，而不是它说它做的事；谁持有管理密钥，它决定分离原则是写在代码里还是写在承诺里——一个能被单个密钥修改税率的协议，分离只存在于文档中；代币的流向，它决定"税"是真的流向了存在凭证的持有者，还是流向了某几个地址。五件事都查完，你对这个协议的了解，就超过了它的任何一篇介绍文章，包括这一篇。

为什么要很多个方案

我在第一篇说过，三重盗窃不是三个坏人干的坏事，是账本的缺口。那么补缺口的事，也不应该由一个方案来做。

理由在分离原则本身。如果全世界只有一个分配协议，它就是唯一的分配者，而唯一的分配者迟早会变成捕获者——这正是分离原则要防止的事。分配协议之间的竞争，本身就是分离的保障：没有单一分配者，就没有人能圈地。

所以这一篇的结尾不是推荐，是邀请。如果你认为这个协议的三层哪一层错了，写下你的版本，用同一把尺子量它，把它放到链上。评价的标准已经在这里了：依据不依赖劳动，捕获与分配分离，同意的三个检验，可以被验证。一个方案不需要在四条上全部完美，但它必须回答全部四条。

下一篇离开协议，回到人。账本可以记录时间，但产生时间的，仍然是一个会死的身体。

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法币偷走你的过去，算法偷走你的现在，AI 偷走你的未来；时间主权，就是把三者夺回来。

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梯度

时间盗窃系列 · 四 · 展开《时间主权宣言》第三节

> 熵增是宇宙的税，秩序是你用梯度买来的东西。你的身体、你的注意力、你的部落，是你仅有的三个梯度。 ——《时间主权宣言》

生命不对抗熵

前三篇讲账本。这一篇讲账本记录的那个东西从哪里来：时间是从一个会死的身体里出来的，而身体是一台热力学机器。

热力学第二定律说，孤立系统的熵只增不减——秩序总是瓦解，热的总会变冷，整齐的总会变乱。这句话被引用得太多，以至于派生出一种流行的悲壮感：既然一切秩序注定瓦解，那么一切努力都是对抗宇宙的徒劳。我在一年前的旧稿里也这样写过，那是错的，而且错得有害，因为它把问题问错了。

1944 年，薛定谔在《生命是什么》里给出了正确的问法。生命不是孤立系统，它是开放系统；它不对抗熵增，它"以负熵为食"——从环境中摄取秩序，把自己产生的熵排出去。三十年后，普里戈金把这件事做成了物理学：在远离平衡的开放系统里，只要维持一个梯度——温差、浓度差、能量差——秩序就会自发出现，他把这种结构叫作耗散结构。一锅被加热的水底部会自发形成整齐的对流元胞，一场飓风是大气在温差之上自组织出来的巨大秩序，一个细胞是化学梯度之上的耗散结构。秩序不是对抗熵增得来的，是消耗梯度买来的。梯度一断，秩序立刻瓦解。

所以正确的问题不是"如何对抗熵增"。熵增是宇宙的税，没有人能免。正确的问题是两个：你的梯度是什么？它是否正在被别人收走？

回头看第一篇，三重盗窃偷的正是三个梯度。法币抽走你用身体换来的储备；算法切碎你的注意力；AI 让你的专业不再足以聚起一个部落。这一篇逐个讲这三个梯度——它们是你仅有的三个。

身体：能量变成时间

身体是一台把能量转换成可用小时的机器。它的效率，是每一单位摄入的能量能换来多少清醒、专注、无痛的小时。这个定义把所有关于健康的讨论都压缩成一个问题：这件事是在增加我的梯度，还是在消耗它？

关于身体的证据，诚实的做法是分级说，因为这个领域充满了被夸大的东西。

力量训练的证据最硬。2022 年，门马等人在《英国运动医学杂志》上发表的荟萃分析汇总了十六项前瞻性研究，发现每周 30 到 60 分钟的肌力训练，与全因死亡风险降低 10% 到 17% 相关，心血管病、癌症、糖尿病的风险也相应降低。这是观察性证据，不是因果证明，但它的一致性和剂量关系都很清楚。肌肉是身体里最大的代谢器官，它的存量决定你的能量转换效率，而它从三十岁起每十年流失几个百分点——不训练，梯度就在匀速流失。

睡眠的机制证据来自 2013 年谢等人在《科学》上的研究：睡眠时，脑脊液对代谢废物的清除效率显著提高。这是小鼠实验，推到人身上需要谨慎，但它第一次给"睡眠是大脑的清理过程"提供了一个物理机制，而不只是一个比喻。大脑是耗散结构里最贵的那一个，它的废物不清，秩序就不能重建。

间歇性禁食的证据在中间。德卡博和马特森 2019 年在《新英格兰医学杂志》的综述总结了动物实验和人类试验：在动物中，限时进食和热量限制延长寿命的证据很强；在人类中，证据集中于代谢指标的改善——血糖、血脂、体重——寿命的证据还没有。大隅良典因阐明自噬机制获得 2016 年诺贝尔奖，这是机制层面的基础，但"禁食十六小时就能激活自噬从而延缓衰老"这种说法，是从机制到结论之间跳了好几步。我在旧稿里把它写成了"细胞层面的熵减"，那是贴标签，不是论证。

所以这一节不给清单。给一个原则，它来自塔勒布：有机体在适度的压力下变强，在持续的舒适里变弱，他把这叫反脆弱。身体的梯度不是靠减少消耗维持的，是靠刺激-恢复的循环维持的——训练是刺激，睡眠是恢复，饥饿是刺激，进食是恢复。舒适取消了刺激，也就取消了循环。舒适是熵增的温床，压力是熵减的扳机，这句话的物理含义就是：梯度需要被反复建立，不是被保护。

体脂率是身体的通胀率——不是审美数字，是储备被稀释的速度。这是旧稿里我愿意保留的一句话。

注意力：时间变成秩序

身体把能量变成小时，注意力决定这些小时能变成什么。一个没有注意力的小时，是一个被支出但没有换回任何秩序的小时。

2010 年，基林斯沃思和吉尔伯特在《科学》上发表了一项研究，用手机随机询问两千两百多人此刻在做什么、在想什么、感觉如何。结果是：人们在清醒时间的 46.9% 里，想的不是正在做的事；而且走神的时候，无论想的是愉快还是不愉快的事，人都更不快乐。这项研究的重要性在于

它把注意力和体验连起来了——注意力的完整度不只决定你的产出，还决定你活得怎么样。一个被切碎的现在，不只是低效的，是不快乐的。

第一篇引过马克 2008 年的研究：一次打断之后，回到原任务平均要二十三分钟，而人们用加速来补偿，代价是压力。两项研究合起来，画出了被算法收走梯度之后的样子：一半的时间在走神，走神的时候不快乐，每一次被拉回来都要二十三分钟，追回来的方式是更紧张。

纽波特把反面的能力叫作深度工作：长时间不被打断、把认知推到极限的专注。他的论点是，这种能力在今天同时变得更稀缺和更值钱——稀缺是因为环境在系统性地破坏它，值钱是因为它越来越难被复制。第一篇讲过，AI 时代最后被定价的东西是判断；判断只在深度注意力里产生。

深度注意力还有一个上限，知道这个上限很重要。埃里克森 1993 年对柏林音乐学院小提琴学生的研究发现，最顶尖的学生每天刻意练习的时间大约是四小时，而且分成几段，段与段之间休息；超过这个量，练习的质量下降，收益消失。后来的研究在不同领域反复得到相近的数字。它的含义是：一个人每天真正能把时间变成秩序的，大约只有三到四个小时，其余的小时是维持、恢复和消耗。这把注意力的梯度算得很清楚——你一天只有四个小时是"有名字"的时间，而算法要拿走的正是这四个小时，因为只有这四个小时值得拍卖。

实践上，注意力的梯度靠四件事维持。时间块：把一天切成几个大块，每块只做一件事，块内关掉一切通知。环境：一个物理上专门用来专注的地方——房间、桌子、哪怕一副耳机——让身体知道进入这里就是进入专注。单任务：并行是一个幻觉，大脑没有并行能力，只有快速切换的能力，而每一次切换都付二十三分钟。外部记忆：把记忆外包给笔记系统，福特把这叫"第二大脑"，它的意义不是记得更多，是把注意力从"记住"里解放出来留给"判断"。

注意力主权有一个简单的测试：你能不能连续九十分钟不看手机。如果不能，这个梯度已经被收走了，而且收走它的人没有问过你。

部落：秩序得以复利

一个人的秩序会随他死去。一群人的秩序可以传下去——语言、技艺、信任、规则，都是在人与人之间复利的秩序。这是第三个梯度，也是最容易被忽略的一个，因为它看起来不像热力学。

邓巴在 1992 年从灵长类的脑容量推算出人类能够维持稳定社会关系的上限：大约一百五十人。这个数字的含义不是"你应该认识一百五十个人"，而是：部落天生是小的。深度连接的容量是有限的，所以它是稀缺的，所以它是一个梯度。十个可以在凌晨打电话的人，比一千个点过赞的人更接近一个部落。

为什么部落是三个梯度里最重要的？因为它是唯一一个机器不能以零边际成本复制的东西。信任不是一个可以被生成的内容，它是重复博弈的结果——阿克塞尔罗德在 1984 年用计算机锦标赛证明，合作在重复博弈中演化出来，条件是博弈要重复足够多次，参与者要记得上一次。信任需要历时的在场：同一个人，在很多个时间点，做了可以被预期的事。这恰恰是机器没有的东西。机器有算力，没有历时性——它不会在时间里"在场"，所以它不能被信任，只能被使用。

第一篇说，AI 让专业不再足以聚起一个部落。回应在这里：聚起部落的从来不是专业，是在场。专业只是在场的一种证明方式，当这种证明方式贬值，在场本身的价值反而显露出来。这和第三篇讲的协议第一层是同一件事在两个层面的表现——链上记录的是"在时间中出现过"，部落记录的也是。

邓巴后来把那一百五十人拆成了几层：五个最亲密的人，十五个可以求助的人，五十个朋友，一百五十个熟人。每往外一层，每个人分到的时间就少一个量级。梯度在最里面的两层——五个和十五个——因为只有那里的在场密度足够让信任复利。外面的一百三十五个人是环境，不是梯度。

部落也有它的失败方式，必须说。一个只会同意你的部落不是梯度，是消耗——它让你的判断失去校正，而判断失去校正的人，和被算法切碎注意力的人一样，正在失去秩序。检验很简单：过去一年里，这群人有没有让你改过主意？没有的话，你拥有的是一个回音室，不是一个部落。

实践上，部落的梯度靠三件事。公开构建：把你在做的事、你的判断、你的失败公开出来，这是在场最可验证的形式，它让对的人找到你。提供价值优先：在任何关系里先问能给什么，而不是能拿什么，因为信任的第一步总是有人先付出一次。深度优于广度：把时间花在十个人身上，不是一千个人身上。

守住梯度，是热力学

把这一篇压缩成一张表：身体把能量变成时间，法币抽走它换来的储备；注意力把时间变成秩序，算法把它切碎拍卖；部落让秩序复利，AI 拆掉了聚集它的理由。三个梯度，三重盗窃，一一对应。

守住梯度不是自我提升的鸡汤，是热力学：没有梯度就没有秩序，没有秩序就没有时间可记。前三篇讲的账本，记录的是时间；这一篇讲的身体、注意力和部落，是时间的来源。来源干涸，账本就是空的。

所以给自己记一本账，每周三个数字就够了：这周做了几次力量训练；这周最长的一段不被打断的专注有多久；这周为那五个或十五个人做了几件具体的事。三个数字连着几周下降，不用等任何人告诉你，你的梯度正在被收走——而收走它的，往往就是第一篇里的那三套系统。

但梯度终究会耗尽。身体会死，注意力会熄灭，部落会散。那么秩序去哪里？这是最后一篇的问题。

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法币偷走你的过去，算法偷走你的现在，AI 偷走你的未来；时间主权，就是把三者夺回来。

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终局

时间盗窃系列 · 五 · 展开《时间主权宣言》第六节

> 终局不是逃离，是延续。肉体的终点由生物学决定，影响力的终点由账本决定。 ——《时间主权宣言》

逃离的叙事，和它的边界

讲"主权个人"的书，几乎都停在同一个地方：逃离。

1997 年，戴维森和里斯-莫格在《主权个人》里预言，信息技术会削弱民族国家对个人的控制，有能力的人将不再被地理锁定——收入来自全球，资产是加密的，国家从主人变成一个可以被挑选的服务商。2022 年，斯里尼瓦桑的《网络国家》把这个预言往前推了一步：先在线上聚起一个有共同价值的社群，再众筹领土，最后寻求外交承认——先有网络，再有国家。

两本书的洞见是真的。国家确实正在变成可选项，边界确实正在变薄。但两本书回答的都是同一个问题：你在哪里。它们没有回答另一个问题：你的时间去哪里。迁徙不解决时间——你把自己搬到哪个管辖区，时间都以同样的速度流走，法币、算法和模型在任何一个管辖区里都一样运转。换一本护照，是换了一个收税的人，不是换了账本。

我在一年前的旧稿里写过"获得第二本护照"这样的建议。这一篇是对它的修正：那是一个关于地理的答案，而问题不在地理。

赫希曼在 1970 年给出过一个更一般的框架：面对一个让你失望的组织，你有两种反应——退出，或者呼吁。退出是用脚投票，呼吁是用声音改变它。主权个人的叙事是纯粹的退出；政治是呼吁。时间主权走的是第三条路：既不退出，也不呼吁，而是换一本账。你不离开系统，也不试图说服系统，你在它旁边建一个记录时间的地方，然后让记录本身说话。前三篇讲的协议，就是这第三条路的一个实例。

所以终局不是逃离。逃离是空间上的动作，而时间主权的终局必须是时间上的：延续。

影响力的终点由账本决定

肉体的终点由生物学决定，这件事暂时没有人能改。但一个人死后，他的思想、判断、作品是否继续起作用——这件事不由生物学决定，由记录的地方决定。

记在哪里，决定能活多久。看三种记法。

记在平台上的，随平台消失。皮尤研究中心在 2024 年做过一项统计：2013 年存在的网页，到 2023 年有 38% 已经无法访问；政府网站和新闻网站的页面也在以相近的速度消失。哈佛法学院图书馆 2021 年的研究查了《纽约时报》文章里的深层链接，四分之一已经失效。这些不是边缘案例，是互联网的常态。每一个关闭的博客平台、停运的社交网络、改版的网站，都带走一批人的记录，而被带走的人从来没有被通知。互联网承诺了记忆，只留下了遗忘。

记在硬盘上的，随硬盘损坏。这件事不需要统计——每个人都丢过至少一块。

记在后人记忆里的，三代以内归零。做过家谱的人都知道这个数字：曾祖父母一辈，多数人连名字都不完整。

所以"影响力的终点由账本决定"不是修辞。它是一个关于存储介质的陈述：你的秩序能延续多久，等于它所在的介质能活多久，而上面三种介质的寿命都短于一个人想要被记住的时间。

有人试过为更长的时间设计。1996 年，布兰德和希利斯发起了 Long Now 基金会，要造一座走一万年的钟，每一百年报时一次。它的意义不在钟本身，而在一个问题：如果你必须为一万年设计，你会怎样做？答案和"为十年设计"完全不同——不能依赖任何一家公司、任何一种电源、任何一代人的记忆。这是档案问题的正确尺度：不是"怎样保存到我死后"，而是"怎样保存到没有人记得我是谁之后"。

永不关闭的档案

有没有一种介质，它的保存不依赖任何机构的善意？

2019 年，威廉姆斯等人在 Arweave 的黄皮书里描述了一种机制：一次付费，永久存储。它的经济模型是一个捐赠基金——你支付的费用里，一小部分用于当下的存储，大部分进入基金，基金的收益用来支付未来的存储成本。这个模型依赖一个假设：存储的成本会持续下降，历史上它每年下降约三成，模型按远低于这个速度的保守假设计算，得出至少两百年的结论。节点保存数据的动机不是善意，是激励：节点必须证明它能访问随机抽取的历史数据，才能获得记账的权利。

必须诚实地说清这个方案的边界。两百年是一个经济假设，不是一个物理保证；网络可能消亡；激励可能失效。但它和前面三种介质有一个本质区别：它是目前唯一一个把"保存"从机构的善意变成了经济激励的方案。平台保存你的东西，是因为它暂时有用；硬盘保存你的东西，是因为它暂时没坏；永久档案保存你的东西，是因为有人因此获利，而且这个获利的结构被设计成不需要任何人决定。这就是"靠数学和经济激励、而不是靠任何机构的善意"的意思。

对一个人来说，这意味着三件事。你的思想、决策、作品可以被刻进一个不随平台消失的地方。刻进去的东西不可篡改——不是因为没有人想改，是因为改的成本高于任何人的动机。它不依赖任何人的恩赐——没有一个账号可以被封禁，没有一个公司可以倒闭。

这份宣言和这五篇文章，先用了同一思路的轻量版本：每份文本的指纹记在比特币区块链上，原文托管在公开的论文库里。不是因为它们重要，是因为这是第六节自己先要做到的事。

数字继承

档案解决的是"记在哪里"。继承解决的是"记下来的东西如何继续起作用"。三件事。

第一，存在凭证的继承。韦尔、奥尔哈弗和布特林在 2022 年的论文里描述了一类不可转让的凭证——灵魂绑定代币——它和一个人绑定，不能买卖，但可以设定恢复和继承的规则：由你指定的人，在你指定的条件下，接管它。第三篇讲的存在凭证，正是这一类东西。它的继承，意味着你在账本上的那一栏不随你消失。

第二，账本的自动结算。智能合约可以在你不在场的时候继续执行你定下的规则——按你的意愿分配、转移、支付，触发条件是一个你不再响应的信号。这不是新技术，是"死者开关"的链上版本：你的判断变成了代码，代码在你之后继续结算。

第三，判断的代理。一个人一生的文字、决策、作品，足以训练出一个在风格和倾向上接近他的模型——它可以继续写、继续评估、继续回答问题。这是第十二章旧稿里最激进的部分，这一篇要给它划清边界。

边界是这样的：模型不是你。它是你的判断的统计影子，在你写过的问题上像你，在你没写过的问题上只是猜。它没有你的梯度——它不会死，所以它不会珍惜，而珍惜恰恰是判断的来源。所以它的权限必须有限，而且这个限制可以直接从分离原则推出来：它可以建议，不能决定；它可以分配，不能捕获。一个能替你捕获价值的代理，迟早会变成一个为自己捕获价值的代理。

还有后人的同意。一个"永不离线的数字祖先"对后代可能是财富，也可能是负担。设计上必须允许后人关闭它。延续不是强加，被记住的权利不包括强迫别人记住。

还有一件没有解决的事：法律。遗产法认得遗嘱、信托和继承人，不认得智能合约和灵魂绑定凭证。一个在链上自动结算的账本，和一份在法院认证的遗嘱，如果说的不一样，目前几乎所有法域都会站在遗嘱一边。所以数字遗嘱不能替代法律遗嘱，只能和它并列，而且两者必须说同一件事——这是现阶段唯一稳妥的做法。它和第三篇的"法律定性"是同一个未解问题的两面：账本在前面走，法律在后面追，中间这段距离由你自己承担。

最后划一条哲学上的线，免得被误读。数字继承不是永生，不是意识上传，也不回答"那个模型是不是我"的问题——忒修斯之船的问题在这里没有答案，也不需要答案。它回答的是一个更小、更实际的问题：一个人的判断和作品，能不能在他死后继续按他定的规则起作用。能。这就够了。

实践上，数字继承始于今天的四件事。统一身份：用一个由你掌控的去中心化身份和存在凭证，标记你的所有权和作品，这是未来继承的入口。结构化记录：把你的思考、决策、感悟写下来，这不只是日记，是未来训练代理的语料，也是后人理解你判断的依据。永久存档：把最重要的作品和文件放进永久档案，从今年开始，不用等到完整。数字遗嘱：写明谁继承你的凭证，代理有什么权限，后人可以在什么条件下关闭它。

不是不死，是不被清零

如果时间是货币，死亡就是最终的破产。这句话的意思不是悲观，是会计：一个人的时间账户在死亡那一刻关闭，此前所有的秩序——判断、作品、关系——如果没有被记在一个能延续的地方，就随账户一起清零。

代码第一次给了我们重组的机会。重组不是不死，破产重组的意思是：债务人不在了，但资产和规则还在，还可以继续按规则运作。不是不死，是不被清零。

五篇到这里收束。第一篇：三套系统偷时间，它们能偷是因为时间在账本上没有名字。第二篇：旧账本只记劳动和持有，机器出时间之后它记不下去了，捕获与分配必须分离。第三篇：有人试着把"存在"写进账本，用尺子量它，它是一个可以被验证的解，不是唯一的解。第四篇：时间从身体、注意力和部落三个梯度里来，守住梯度是热力学。第五篇：梯度终会耗尽，但记在能延续的账本上的秩序不会。

宣言的最后一句是：从今天起，记在你自己的账上。读完五篇，这句话应该有了具体的含义——记在一个记录存在的账本上，记在一个不依赖善意的介质上，记在一个你死后仍然按你的规则结算的地方。

那就是你的账。

— • —

法币偷走你的过去，算法偷走你的现在，AI 偷走你的未来；时间主权，就是把三者夺回来。

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The Sovereignty of Time

A Declaration · Complete Edition

Titan · v1.1 · November 2026

ONE SENTENCE

Fiat steals your past, algorithms steal your present, AI steals your future. Time sovereignty is taking all three back.

ONE PARAGRAPH

Time is the only non-renewable resource. Three systems have learned to steal it: fiat dilutes the savings you traded your time for, algorithms shred the attention you have right now, and AI is taking your pricing power over the time you have left—and none of them asked for your consent. After AI, production is no longer scarce; distribution is. Time sovereignty is not an exit. It is recording existence itself on a ledger that cannot be diluted, and making every output amplified by machines pay a tax to human time. This is not the only solution. It is one that can be verified.

I. Time

Time is the only non-renewable resource. Money can be earned again, information can be copied, energy can be converted. Time moves in one direction only. This is not rhetoric; it is physics. Entropy gives time its arrow, and you happen to live along it.

So the foundation of every sovereignty is sovereignty over time. Territorial sovereignty, monetary sovereignty, data sovereignty — each answers the same question: who gets to decide what happens during a given stretch of time. For an individual the question is simpler and harsher. Your life is a sum of time being spent, irreversibly, and you have never seen the ledger.

This declaration does one thing. It opens that ledger: where your time has been going, why you never agreed to it, and how to record it back under your own name.

II. Three Thefts

Three systems, one after another, learned how to steal time.

By “steal” I mean something specific: transfer without consent. Not robbery — you would resist robbery. Not fraud — you would discover fraud afterwards. Theft, in this sense, happens inside the default settings. You do nothing, and it is done.

Fiat steals your past. Savings are the condensed form of time you have already spent: years of early mornings and late nights, exchanged for a number. That number stands for time, not paper. And the one real freedom a fiat system has is to issue more. Every issuance leaves the number in your account untouched and changes only how much time it can buy back. Inflation is not an economic phenomenon. It is the theft of time from savers — the gentlest possible way to take the least renewable thing you own.

Algorithms steal your present. Attention is what time looks like right now: whatever you are looking at is what you are living in. The business of a recommendation engine is to cut that time into pieces, auction them, and cut again. Every refresh is a bid, and what is being bid on is the next fifteen seconds of your life. You are not a party to the auction. You are the lot.

AI steals your future. Expertise used to be your pricing power over the time you have left: ten years of study is why your hour is worth what it is. When cognition can be copied without limit, and each copy costs almost nothing, that pricing power leaves your hands. This does

not mean you will lose your job. It means your future hours will be priced by someone else — and priced cheaply.

The three have one thing in common: none of them asked you. You never signed an agreement letting a central bank dilute your savings, or an algorithm auction your attention, or a machine train on your output and then price you. Staying in requires no action. Leaving each one does. That is what theft means.

III. Gradient

The iron law of the universe is that entropy increases. The sentence has been repeated so often that most people draw the wrong conclusion from it: since order is doomed to decay, every effort is a heroic, losing struggle.

That is not how life has ever survived. Life is an open system. It does not fight entropy; it consumes an energy gradient, exports its entropy, and buys internal order with the difference. The temperature gap between the sun and the earth feeds everything alive; the gap between a bowl of rice and a hungry body feeds you. Entropy is the universe’s tax. Order is what you buy with a gradient. No one is exempt from the tax, but everyone decides who gets their gradient.

So the question for a sovereign of time was never “how do I fight.” It is two more concrete questions: What is your gradient? And is someone else collecting it?

You have exactly three. Your body, where energy becomes time — its efficiency sets how many hours you have. Your attention, where time becomes order — its integrity sets what those hours can produce. Your tribe, where order compounds — one person’s order dies with them; a group’s order can be passed on. The three thefts take precisely these three gradients: fiat drains the reserves your body earned, algorithms shred your attention, and AI makes your expertise no longer enough to gather a tribe around.

Guarding your gradients is not self-improvement advice. It is thermodynamics.

IV. Distribution

The old mechanism of distribution was labor: you put in time, you got back a share. That chain ran for ten thousand years. It was unfair, but it was legible — how much a person held was, roughly, a function of how much time they had put in. Wages, working hours, pensions, labor law: all of it is patchwork on that one chain.

When machines put in the time, the chain breaks. A machine can supply ten thousand hours of cognitive labor in a second, and none of those hours belong to anyone’ s life. After AI, production is no longer scarce. Distribution is. Machines amplify every output, and no mechanism says who the amplified part belongs to — so by default it belongs to whoever owns the machine.

A civilization can therefore have unlimited output and nowhere to put its people. This is not a technical problem; the technology has already solved production. It is a ledger problem. Our ledgers only know how to record “who did the work.” They have no entry for “who exists.” When labor stops being the basis of distribution, the ledger has to find another one.

Here is a principle I take to be a hard constraint: the mechanism that captures value cannot also be the mechanism that distributes it. When the owner of the machine decides who gets the machine’ s output, that is not distribution. It is enclosure. Capture and allocation must be separated. It is the ruler I hold against any proposed scheme of distribution.

V. A Protocol

Changing the currency is not enough. Bitcoin changed who may issue money; memecoins removed the barrier to issuing it; stablecoins simply moved fiat itself onto the chain. Each succeeded on some axis, and none escaped fiat: as long as a currency’ s rise and fall is decided by capital flows, its value is still denominated in fiat and its holders are still playing a casino’ s game. The revolution is incomplete — not for lack of technology, but because they changed the unit on the ledger without changing what the ledger records. It still records “who holds how much,” not “who exists.”

Someone has already written a different idea into a protocol and has been testing it on several chains since 2023. I treat it as one of many possible schemes of distribution. Its design has three layers.

First, proof of existence in time. Existence itself is recorded as a credential that cannot be diluted. Existence is not identity, not an asset, not what you own or what you have done; it is the fact that you were here, in time. It is the hardest fact in the off-chain world, and the only one a machine cannot generate on your behalf. Put it on the ledger, and distribution has a basis that does not depend on labor. This layer can be implemented many ways; this protocol is one.

Second, a market for attention and creativity. On top of the existence credential, human attention and human creativity are priced — by the people giving them, not by the platform. This layer turns the present that algorithms stole into a present you can sell yourself.

Third, an AI production tax. Every output amplified by machines pays a tax to human time. Not to a government: to the holders of existence credentials — to everyone who has been here in time. The value a machine creates with ten thousand hours of cognitive labor flows back to the people the machine has priced.

It has run into the problems every protocol of this kind must face: how proof of existence resists fabricated existence, who sets the tax rate, how a market prices anything without liquidity. These are not all solved. It has also proven something possible: a basis for distribution that does not depend on labor can be written as code, used by a few hundred strangers, and stay alive across several chains for nearly four years.

It is not the only solution, and it may not be the best one. It is a solution that can be verified. I would rather see you propose your own protocol of distribution than believe in this one.

VI. Endgame

The endgame is not escape. It is continuation.

The sovereign-individual story usually stops at escape: change your country, change your currency, change your identity. That mistakes sovereignty over time for freedom of movement. Moving does not solve time. Wherever you take yourself, it runs out at the same speed.

The end of the body is set by biology. The end of influence is set by the ledger. Whether a person’s thoughts, decisions and work keep acting after they die depends on where those things are recorded. Recorded on a platform, they vanish with the platform; on a hard drive, with the drive; in the memory of descendants, within three generations. The internet promised memory and delivered forgetting.

Carve them instead into an archive that never closes — one kept alive by mathematics and economic incentive, not by any institution’s goodwill. Let them go on being read, executed and inherited after you: your judgment trained into an agent that keeps judging in your place, your existence credential assigned to an heir, your ledger settling while you are no longer present.

If time is money, death is the final bankruptcy. Code has given us, for the first time, a chance at reorganization — not immortality, but not being zeroed out.

VII. Choice

You cannot choose whether time runs out. Physics decided that for you.

You can only choose whose ledger it is recorded on. On the central bank’s ledger it is inflation. On the platform’s ledger it is traffic. On the machine owner’s ledger it is depreciation of compute. On your own ledger, it is your life.

This declaration asks you to believe nothing. It asks you to look at the ledger, once.

From today, record it on your own.

Three Thefts

The Theft of Time • Part 1 • Expanding Section II of the Declaration

> *Three systems, one after another, learned how to steal time. The three have one thing in common: none of them asked you. — The Sovereignty of Time*

Why "theft"

The word is heavy, so it needs defining first.

Robbery is overt. Someone takes what is yours in front of you; you know what happened, you can resist, and the law is on your side. Fraud is delayed. You agreed at the time, then found the premise was false, and you can seek redress. Theft, in the sense used here, is neither. It happens while you are not in the room, inside the default settings. You do nothing, and it is done; you notice nothing, because it was designed from the start not to be noticed.

I use "theft" rather than a politer word because every politer word hands the responsibility back to you. "Inflation is part of the cycle" — so you should learn to invest. "Scrolling is a self-control problem" — so you should be more disciplined. "Being replaced by AI means you stopped learning" — so you should reskill. The three sentences have the same structure: the system takes something from you, then tells you it is your fault.

"Theft" puts the responsibility back on the system. The definition is specific: transfer without consent. The three sections below take one system each — how it performs the transfer, which part of your time it takes, and why you were never asked.

I. Fiat steals your past

Start with what savings are.

Savings are not money. Savings are the balance of time you have already spent. In some year, on some few hundred early mornings and late nights, you did things other people needed and received a number in return; the number stands for those mornings and nights, which are gone and cannot be had again. Savings are in the past tense. They are the part of your life that has been spent and not yet redeemed.

On 15 August 1971 the United States closed the window at which dollars could be exchanged for gold. From that day, the quantity of the world's major currencies was no longer bound by anything physical, only by policy. The significance is not gold itself — gold was merely the carrier of a constraint — but what the system gained once the constraint was gone: the freedom to issue.

How does issuance steal? It never touches the number in your account. Your hundred is still a hundred. What changes is how many hours of other people's labor that hundred can buy back. By the US Bureau of Labor Statistics' consumer price index, 100 dollars in 1971 corresponds to roughly 832 dollars in 2026 — which is to say that a person who put away 100 dollars in cash in 1971 and held it until today holds about 12 percent of the purchasing power they started with. An average of 3.9 percent a year sounds mild; after fifty-five years, an eighth is left. Every fiat currency does this; they differ only in speed.

Economics has a word for it: seigniorage. The issuer creates new money at almost no cost and uses it to buy real goods and labor. The tax base is everyone holding the currency, the rate is the pace of issuance, and the method of collection is dilution. It appears on no tax return.

The tax has a second property: it is regressive. New money does not fall evenly on everyone. It enters at the point of issue and moves outward — through banks, governments, holders of large assets — reaching wages and cash savings last. Those who get it first spend at old prices; those who get it last spend at new ones. This is the Cantillon effect. Asset holders are compensated for inflation because assets rise with the money supply; people who live on wages and cash savings are not, because wages are always the last to rise. Seigniorage is therefore not only a tax without a return; it is a tax paid mostly by the patient and the asset-less.

There is a subtler layer. Irving Fisher, in 1930, explained interest as the reward for waiting: you postpone consumption, you let others use the resources first, and interest compensates your patience. When the real rate is negative — when issuance outruns interest — waiting is no longer compensated but punished. A system that persistently punishes patience and rewards immediate consumption and debt is stealing more than the purchasing power of savings. It is stealing the meaning of saving. It teaches, systematically, not to set anything aside for the future.

The counterargument deserves a hearing: moderate inflation is necessary, and deflation is worse — falling prices make people delay purchases, revenues fall, layoffs follow, prices fall further. At the macro level the argument has merit, and I will not rebut it here. But it answers a different question. It answers "what the system needs," not "who pays." Even if moderate inflation is optimal for the system as a whole, that optimum is achieved through a tax that no one consented to, no one sees, and savers mostly pay. A necessary tax is still a tax, and a tax without a return still needs someone to ask: who agreed?

No one asked you. A saver's "consent" is presumed from the act of holding the currency, and you were born into a fiat jurisdiction — holding it was not a choice but a starting point. Your only exit is to exchange it for something else, and what to exchange it for, when, and what happens if you are wrong, are costs and risks borne entirely by you. Staying requires no action; leaving requires knowledge, judgment and luck. That is the power of the default.

One more thing happened in 1971: Nicholas Georgescu-Roegen published *The Entropy Law and the Economic Process*, arguing that economic activity is fundamentally an entropic process — low-entropy resources in, high-entropy waste out, irreversibly. That connects to the deeper frame of this series and is taken up later. But the coincidence of dates is not nothing. People of that era already sensed that once a constraint is removed, some things only flow one way.

II. Algorithms steal your present

Also in 1971, Herbert Simon wrote a sentence that has been quoted ever since: a wealth of information creates a poverty of attention. His point was about design — since attention is scarce, information systems should be designed to conserve it. Fifty years later, information systems are designed for the opposite purpose: to consume as much of it as possible.

What is attention? It is what time looks like right now. Whatever you are looking at is what you are living in. The past has been spent, the future has not arrived, and the only thing you truly possess is the second passing now — and the content of that second is wherever your attention is. To steal attention is not to steal some portion of your time. It is to steal the only tense in which you are alive.

How is it stolen? By auction.

The core mechanism of internet advertising is a real-time auction. In 2007, Edelman, Ostrovsky and Schwarz described in the *American Economic Review* the "generalized second-price auction" used by Google and Yahoo: every ad slot is auctioned at every impression; the highest bidder wins and pays the second-highest price. The mechanisms have grown more complex since, but the skeleton stands. Every time you refresh a page, an auction completes in tens of milliseconds, and what is being sold is "this person's attention for the next few seconds."

Look at who is in the room. The buyers are advertisers; the seller is the platform; the auctioneer is the algorithm. What are you? You are the lot. The lot does not set the price, does not share the proceeds, and does not know the auction took place.

This makes the recommendation engine's role clear. Its objective function is not "find this person the best content" but "keep this person here as long as possible," because time on site determines the number of auctions. It is not finding content for you; it is finding you for advertisers. The algorithm seems to understand you because understanding you is the shortest path to keeping you.

So no detail of the product is neutral. Infinite scroll removes "reaching the bottom," a natural stopping cue. Autoplay removes "pressing next," a decision. Red badges and vibrations convert an action you used to initiate — "let me see if there's anything new" — into one initiated by the device. None of these designs exists to help you decide when to stop; all of them exist to remove "stop" as an available option. The length of time a person is auctioned equals the length of time they cannot find the stop button.

The cost is measurable. In 2008, Gloria Mark and colleagues at the University of California, Irvine, interrupted people at work and measured how long it took them to return to the original task: about twenty-three minutes on average. Her second finding matters more: people compensate for interruptions by working faster, at the cost of more stress and frustration. Fragmentation does not steal only the minutes of the interruption; it steals the twenty-three minutes afterward, and the extra toll you pay trying to win them back.

The counterargument: the service is free, so this is an exchange, not a theft. It fails twice. First, the price is not zero — you pay in the only non-renewable thing you own; the bill simply omits that line. Second, an exchange requires both sides to price, and here only one does: the platform decides what an hour of your attention is worth in advertising, and you neither know the figure nor can negotiate it. An exchange in which only one party knows the price is not called an exchange in any market.

No one asked you. What you "agreed" to is a terms-of-service document no one reads, and the default outcome of agreeing is that your attention is auctioned, the proceeds go to the platform, and you receive free content — which is to say, you think you are consuming, and in fact you are being consumed. Leaving requires uninstalling, self-restraint, and the social cost of absence. Staying requires nothing.

III. AI steals your future

What is expertise? It is your pricing power over the time you have left.

A person spends ten years learning a profession not for those ten years but for the thirty that follow, in which an hour of their time sells for more than someone else's. That price is their valuation of their own future hours, and its basis is: others cannot do this, or cannot do it as well. Expertise is a contract about the future, bought with ten years of the past.

Now look at what happened to the other party to that contract.

In November 2024, an analysis by a16z tracked the price of large language model inference: for models of equivalent capability, the cost per million tokens fell from 60 dollars to 0.06 dollars over three years — a factor of a thousand, or roughly tenfold a year. That pace exceeds the decline of computing costs in the personal-computer era. The specific numbers change monthly; the direction does not. The marginal cost of cognitive labor is heading toward zero.

Goldman Sachs estimated in March 2023 that generative AI could expose the equivalent of 300 million full-time jobs to automation, with office and administrative work most affected — precisely the jobs priced on expertise.

But unemployment is not the heart of this. Most people will not be told one day that they have been replaced. What happens is quieter: your hour gets repriced. The new basis is no longer "others cannot do this" but "what does it cost a machine to do this." The future contract you bought with ten years of your past has been unilaterally revalued, and the one doing the valuing is not you.

The counterargument is historical: every technological revolution destroyed some jobs and created others — steam, electricity, computing — and this one will be no exception. I hope that is right, but one structural difference cannot be skipped. Every previous substitution replaced what humans sold cheaply — muscle, repetition, memory — and pushed people toward what sold dearly: judgment, creativity, expertise. Every previous substitution raised the price of cognition. This one substitutes cognition itself. The mechanism "pushed toward something dearer" requires that something dearer exists; this time, we have not yet seen where that next rung is. Perhaps it will appear. Until it does, the people being repriced have nowhere to go, and the repricing has already happened.

This is the theft of the future. Fiat takes the time you have already spent; algorithms take the time you are spending; AI takes the time you have not yet spent but have already priced.

There is a deeper layer still. These models are trained on human output — articles, code, conversations, work — including yours. Your output is taken first to train the model, and the trained model then returns to price you. Two transfers, neither with your consent. The first takes your past output; the second takes your future pricing power. And "opting out," as with the other two thefts, is an option you must discover, apply for, and prove — the default is in.

This is not a moral accusation. It is a structural description: at every link in this chain, your time is not a variable.

IV. None of them asked you

Put the three together and the structure is identical.

Each theft is a default. Staying requires no action: you are born into fiat, you open your phone into the algorithm, your output is in the training data the moment it is published. Leaving each one requires an action, and the cost of that action — knowledge, time, risk, social price — is borne entirely by you.

Each theft hands responsibility back to you. Inflation teaches you to invest, the algorithm teaches you self-discipline, AI teaches you lifelong learning. All three are true, and none is the point. The point is that in the objective function of each of the three systems, your time is not a term. The central bank's target is prices and employment; the platform's is time on site; the model company's is capability and cost. No one wants to steal your time. It simply is not on anyone's ledger.

So what would consent look like? Three tests, each of them plain:

First, you can see it. When the transfer happens, you can see the amount, the time, and the destination. Seigniorage has no return; the auction does not notify the lot; training does not notify the author. All three fail.

Second, you can say no, and saying no costs nothing. An option that requires money, time or risk to refuse is not an option; it is a tollbooth. Changing currency, quitting the phone, applying to opt out of training — all three charge a toll.

Third, your time is a variable on the ledger. Only when a transfer is recorded on a ledger denominated in your time can "consent" even be defined — otherwise you cannot say what you would be consenting to. On the ledgers of all three systems, that entry does not exist.

The three thefts are not three bad acts by three bad actors. They are three expressions of the same gap: our ledgers record "who holds how much" and "who did the work," and not "who exists" or "whose time." Money can be issued without limit, attention can be auctioned without paying its owner, and cognition can be copied without paying its author, because of one shared precondition — time has no name on the ledger.

So the question is not "how do we resist." Resisting a system that does not treat you as a variable is like protesting to a machine: it cannot hear you, not out of malice, but because it has no input port for that.

The question is: what should the ledger record?

That is the subject of Part 2. When machines begin to supply time, labor ceases to be the basis of distribution — and on the old ledger, besides labor, there is nothing.

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Fiat steals your past, algorithms steal your present, AI steals your future. Time sovereignty is taking all three back.

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Distribution

The Theft of Time • Part 2 • Expanding Section IV of the Declaration

> *The old mechanism of distribution was labor: you put in time, you got back a share. When machines put in the time, the chain breaks. — The Sovereignty of Time*

A ten-thousand-year chain

Part 1 ended with a conclusion: the three thefts are possible because time has no name on the ledger. This part is about what the ledger does record, why it is about to stop working, and why changing the currency will not save it.

Humanity has had only one large-scale mechanism of distribution: labor. You put in time, you got back a share. Among hunter-gatherers, sharing was the norm, but the basis of sharing was still who had contributed. Agriculture fixed the chain in place — land plus labor, for ten thousand years. Industry refined it to its barest form, the hourly wage: what you sell is not a product but hours. The chain has one remarkable virtue: it is legible. It is not fair — capital, land and birth keep separate accounts outside it — but how much a person holds is, roughly, a function of how much time they have put in. Every struggle over fairness in the last two centuries — the eight-hour day, the minimum wage, pensions, unions, labor law — has been patchwork on this chain, never an attempt to replace it.

In 1930, Keynes wrote a short essay, "Economic Possibilities for our Grandchildren." He predicted that within a century technological progress would solve the problem of production, that people would need to work only fifteen hours a week, and that the remaining difficulty would be what to do with leisure. He saw half of it. Production is indeed being solved — output per person has multiplied several times since he wrote. But the fifteen-hour week never arrived, and the reason is not technology but the ledger: shares are tied to labor, no work means no share, so no one dares work less. Keynes assumed distribution would follow production automatically. It did not, because the chain itself needed replacing, and he did not see that.

The chain was fraying before AI. Karabarbounis and Neiman, in 2014, measured the global labor share — the fraction of national income going to labor — and found it had fallen by about five percentage points since the early 1980s, across countries and industries. Another set of figures is more vivid: the Economic Policy Institute's long-running "productivity–pay gap" shows US labor productivity rising about 80 percent since 1979 while the hourly compensation of the typical worker rose about 30 percent — for the thirty years before that, the two lines had moved together. Output and share parted ways in the late 1970s, a few

years after the monetary constraint described in Part 1 was removed. Capital has been quietly substituting for labor for four decades. AI is not the start of the trend; it is the accelerator.

When machines put in the time

A machine can supply ten thousand hours of cognitive labor in a second, and none of those hours belongs to anyone's life. That sentence contains something that has never existed before: time without a life behind it.

The labor chain works because time comes out of a life — behind every hour of labor there is a person who lived that hour once and will not live it again, and the share compensates that expenditure. When time can come from somewhere other than a life, the object of compensation disappears. The machine puts in the time; who gets the share? By the logic of the labor chain, the machine's owner, because he "put in" the machine. But what he put in was capital, not time — and the capital chain was never a distribution chain. It is a concentration chain: returns to capital go to capital, by definition.

Korinek and Stiglitz, in a 2017 paper, put the matter plainly: the effect of AI on income distribution is determined not by the technology but by institutions; under present institutions, the likely outcome is a further decline in the labor share and wages pushed toward the cost of machines. They propose various redistributive tools, but every tool is a transfer on the old ledger — let the chain break, then use taxes to carry some share back.

Harari calls the people thrown off this chain the "useless class." The phrase is accurate and cruel, and the way it is accurate deserves attention: it looks at people from the standpoint of the labor chain. A person with no entry on the labor ledger is "useless." Switch ledgers and the same person is not useless; they exist. "Useless" is not a judgment about the person but about the ledger — it says that this book accepts only one kind of input.

So a combination appears that has never existed: unlimited output, and people with nowhere to go. Content can be generated without limit at zero marginal cost; routine cognitive services — most of law, accounting, programming, translation — are becoming a utility. Production is no longer scarce. Distribution is: who gets these outputs, and on what basis. And our ledgers record only two things — who holds how much (capital) and who did the work (labor). Capital is concentrating, labor is depreciating, and there is no third column.

This is not a technical problem. Technology has solved production; it has solved it too well. It is a ledger problem: when labor stops being the basis of distribution, the ledger must find another basis, and technology will not do that on its own.

Some will say: then let everyone own the machines. Employee ownership, universal basic capital, sovereign wealth funds — Norway's oil fund, owned by all its citizens, is the ready example. This is better than transfers, because it reaches the source of capture. But notice what it changes: it admits more people into the "who holds how much" column; the basis of distribution is still holding. Holding concentrates — that is the nature of capital, not a moral flaw — so distribution by holding returns, over time, to where it started, unless holding is bound to something that does not concentrate. What does not concentrate? A person can exist only once. That is why the basis must move to existence, not to "more people holding."

Ledgers have not entirely ignored existence. Censuses record it; citizenship records it; the basic-income experiments of the past decade — Finland's two-thousand-person trial in 2017–18, Stockton's one-hundred-and-twenty-five-person trial from 2019, GiveDirectly's long-running trial in Kenya — all distribute shares by existence rather than labor, and their results broadly agree: incomes stabilized, employment did not fall, mental health improved. They show that distribution by existence works at the level of the person. Their limit is on the other side: where the money comes from. Every one of them was funded by a capturer — a government or a foundation — so they are transfers, not mechanisms; when the funder stops, the share stops. Which raises the question that follows: if the basis is to be existence, the money must come directly from the source of capture, not pass through the capturer's hands.

Why changing the currency is not enough

By this point many readers will have a ready answer: change the currency. Over the past fifteen years three attempts have appeared, each worth taking seriously, each insufficient.

Bitcoin changed who may issue money. Nakamoto's 2008 paper gave the world a currency with fixed supply that no one can inflate — the most direct response to the first theft: savings can no longer be diluted. It did what it promised. But look at what it changed and what it did not. It replaced the unit on the ledger; what the ledger records is unchanged — still "who holds how much." Bitcoin's basis of distribution is hashpower and early arrival, a mixture of the capital chain and timing, not a new basis. It is a better savings technology, not a different distribution.

Stablecoins simply moved fiat itself onto the chain. By 2026 the market exceeds three hundred billion dollars. It proves the efficiency of on-chain settlement, and it proves something more interesting: when people are free to choose their on-chain unit, most choose the dollar. Stablecoins are a victory for efficiency and a surrender of sovereignty — they did not even change the unit.

Memecoins removed the barrier to issuing money. Anyone can create a currency in minutes, which democratizes "the right to issue" to the point of absurdity: millions of micro-central-banks at once. The experiment has an unexpected pedagogical value — it proves that freedom to issue is not justice in distribution. Everyone can mint; the rules of distribution are exactly what they were a century ago: first in, first paid, fastest out wins.

The three share one property: as long as a currency's rise and fall is decided by capital flows, its value is still denominated in fiat and its holders are still playing a casino's game — zero-sum, volatile, attention-driven. The revolution is incomplete not for want of technology but because they changed the unit on the ledger without changing what the ledger records.

Bitcoin's believers will object: a savings technology does not need to solve distribution; it needs to do one thing well. The objection is correct, and it is precisely this series' argument — a savings technology addresses the first theft; it does not address the second or the third, nor distribution after AI. It is necessary, not sufficient. A ledger that can see only holders sees not people but balances; on such a ledger, not holding is the same as not existing.

Capture and allocation must be separated

Now the central principle of this series can be stated. Two definitions first.

Capture is turning output into disposable value — who receives the revenue. Allocation is deciding whom the value belongs to. On the labor chain, the two are performed by a single act: a wage is both capture (the firm pays) and allocation (the worker receives); the market executes them together, so no one needs to tell them apart.

The machine age pulls them apart. Capture concentrates in the owners of machines — that is the nature of capital, and there is nothing wrong with it. The problem is allocation: if allocation is also decided by the machine owners, then everything the machines produce belongs by default to the machine owners, and everyone else receives what the owners choose to give. That is not distribution. It is enclosure. In sixteenth- to eighteenth-century England, the commons were enclosed as private property, and the people who had lived from them became landless laborers — Polanyi called this the origin of labor's commodification. The commons being enclosed today is cognition: models are trained on the output of all humanity, and the output belongs to those who trained the models.

So the principle is: the mechanism that captures value must not also be the mechanism that allocates it.

This is not a new intuition. It is the intuition behind humanity's most mature institutional designs. Montesquieu's separation of powers: those who make the law must not enforce it. Berle and Means' 1932 separation of ownership and control: those who fund and those who

manage must not be one will. Central-bank independence: the department that spends must not be the department that prints. Auditor independence: those who keep the books must not audit them. Each says the same thing — the referee cannot also be a player.

The principle has a corollary that is easy to miss: the state is a capturer too. Taxation is capture. So "let the state allocate" does not automatically satisfy separation — it moves allocation from one capturer to another. True separation requires an allocation mechanism independent of every capturer, firms and states alike. That is why what follows is a discussion of protocols rather than policies: policy is a capturer's instrument; a protocol need not be.

Put this part together with the last, and you have a ruler. Any proposal claiming to solve "distribution after AI" can be measured on four marks:

First, is its basis of distribution independent of labor? If it still distributes by labor, it is patching a chain that has already broken.

Second, are capture and allocation separated? If the allocator is also a capturer — firm or state — it is a refinement of enclosure.

Third, does it pass the three tests of consent: the transfer is visible, saying no costs nothing, and time is a variable on the ledger?

Fourth, can it be verified? An unverifiable scheme of distribution is a promise, not a mechanism.

Part 3 is about an attempt someone has already made to write these four into a protocol. I will measure it with this ruler rather than speak on its behalf.

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Fiat steals your past, algorithms steal your present, AI steals your future. Time sovereignty is taking all three back.

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A Protocol

The Theft of Time • Part 3 • Expanding Section V of the Declaration

> *Someone has already written a different idea into a protocol and has been testing it on several chains since 2023. I treat it as one of many possible schemes of distribution. Its design has three layers. — The Sovereignty of Time*

A note on standpoint

Part 2 ended with a ruler: a basis of distribution independent of labor; capture and allocation separated; the three tests of consent; verifiability. This part applies it to one concrete thing.

I am not this protocol's spokesman and will not speak for it. I treat it as one of many possible schemes of distribution. I chose it to measure because it is one of the few attempts I know of that is already running on-chain with "existence," rather than labor or holding, as the basis of distribution. Once measured, you will see where it stands and where it is still only a statement. The outcome I would most like from this part is that you design a better one.

Its design has three layers. Take them in order.

Layer one: proof of existence in time

For "who exists" to enter the ledger, two questions have to be answered first: how do you prove an existence is real, and how do you keep "existence" from becoming "identity"?

The first question has a name in cryptography: the Sybil attack — one person pretending to be a thousand. When Borge and colleagues proposed "proof of personhood" in 2017, the goal was exactly this: one person, one share, without a central identity authority. Three main routes have since emerged. Biometrics: use the iris or another physical trait to guarantee uniqueness, at the cost of a centralized collector and a key that cannot be replaced once leaked. Social graphs: let already-verified people vouch for one another, at the cost of collusion — a group can vouch non-existent people into existence. Liveness tests: require everyone to complete some machine-resistant task at the same moment, at the cost of tests failing one by one as machines improve. None of the three solves the problem completely. This is unsolved everywhere; it is not one protocol's failure.

The second question — existence is not identity — has a precise tool. In 1985, Goldwasser, Micali and Rackoff introduced zero-knowledge proofs: you can prove a statement true without revealing anything beyond the statement itself. Applied here: you can prove "I am a real existing person, and not any person already registered," without revealing who you are. Existence is recorded; identity is protected. This is not a theoretical possibility; it is technology in use.

The real design choice in this layer lies in the words "in time." Most proof-of-personhood schemes treat existence as a boolean — person or not — and then give one vote or one share per person. This protocol's credential accumulates over time: each period you are present, the ledger gains an entry. So "existence" turns from a yes/no question into a time series: the ledger records not "this is a person" but "this person was present during these periods." Time becomes the unit. That corresponds to the demand of Part 1 — time finally has a name on the ledger.

Its handling of the Sybil problem, as I understand it, is not a claim of solution but an engineering trade: accept that proof of existence cannot be perfect, then use the economics of the two layers above to make fabricated existence unprofitable — as long as the cost of faking one existence exceeds the share that existence can collect, faking loses money. That is a reasonable trade, not a mathematical proof. Whether it holds depends on the parameters of the other two layers, and parameters change. This is the first thing to watch.

Two confusions need clearing. It is not basic income. Basic income is a transfer: a capturer — a government, a foundation, or a company exchanging irises for tokens — puts up the money and pays per head; when the funder stops, the share stops. The existence credential is not a sum of money; it is a column on the ledger. Where the money comes from is the business of layer three; layer one only ensures that "who exists" gets recorded. Nor is it one-person-one-vote. Voting uses existence as a boolean; a credential that accumulates over time records existence as a quantity — how long you were present is how much is recorded. The former answers "are you a person"; the latter answers "where is your time." Those two distinctions set it apart from every proof-of-personhood project of the past decade: those projects prove people; this protocol records time.

Layer two: a market for attention and creativity

Part 1 described an auction: buyers are advertisers, the seller is the platform, you are the lot. Layer two is still a market, with the parties changed: the seller is the person.

This idea has a lineage too. Lanier argued in *Who Owns the Future?* (2013) that every use of a person's data should earn its owner a micropayment. In 2018, Arrieta-Ibarra, Goff, Jiménez-Hernández, Lanier and Weyl sharpened it in the *AEA Papers and Proceedings* into a

question: should we treat data as labor? If the raw material of machine learning is human output, then the people supplying it are workers and should be priced. In 2017, the Brave browser's Basic Attention Token made a partial implementation: users who viewed ads received a share of the revenue. It proved feasibility and exposed the limit — it remained inside the advertising frame, brokered by a platform, with users receiving a split the platform set.

This layer makes three design choices. Pricing power sits with the seller, not the platform — you decide what an hour of your attention is worth, and buyers take it or leave it. Only holders of existence credentials may sell — existence is the ticket of admission, and this is also layer one protecting layer two: machines cannot manufacture attention, because machines have no credential. Creativity and attention enter the same market — a work and a presence are two forms of the same thing, time.

Its problem is equally clear: a market needs buyers. A market with no buyers is a price tag. Who buys human attention and creativity? If the answer is "model companies that need human data to train on," then this layer and layer three are connected — the tax and the market are the same money flowing two ways. If there are no buyers, this layer is only a design. This is the second thing to watch.

Layer three: an AI production tax

"Tax the machines" is not new. Gates proposed a robot tax in 2017: a robot that replaces a worker should pay taxes like one. Korinek has proposed an "AI dividend." California discussed a "data dividend" in 2019. All share one intuition: part of what machines amplify is public — the training data is the output of all humanity; no company created it — so the amplified part should not belong entirely to the amplifier.

One precedent deserves a look, because it has run for more than forty years: the Alaska Permanent Fund. Since 1982, Alaska has placed part of its oil revenue in a fund and paid every resident an annual dividend — one thousand dollars per person in 2025. Its logic matches this layer exactly: oil is value no one created, so it is distributed by existence rather than labor. Alaska's basis for the dividend is simply "you lived here for a year" — an administrative version of proof of existence in time.

This protocol's third layer differs from all of the above in two respects. First, the tax is not paid to a government; it is paid to holders of existence credentials. The reason is in Part 2: the state is a capturer too, and a tax that passes through the state has passed through a capturer's hands, so separation fails. Second, the base is "output amplified by machines," not the machines themselves — which sidesteps the robot tax's definitional problem (what counts as a robot?) at the price of another: what counts as "amplified by a machine"?

This layer has the most unsolved problems. Who sets the rate is a governance problem; how the base is measured is a technical problem; how an on-chain tax reaches off-chain output is a jurisdictional problem. To be plain: layer three is at present more a statement than a mechanism. It says where the money should come from and where it should go; it does not yet say how it is collected. This is the third thing to watch.

Nearly four years of experiment

The protocol has run on several chains since 2023 and has been used by a few hundred strangers. What nearly four years prove and do not prove must be kept apart.

It has proven: that a basis of distribution independent of labor can be written as code; that it can be used by people who do not know one another; that it can survive nearly four years on several chains without depending on any institution. Before 2023 all three were conjectures.

It has not proven: scale — a few hundred people are not a few hundred million; Sybil resistance — no one has seriously attacked it, which is not the same as its withstanding attack; liquidity — nearly four years have not answered how a market prices when no buyers are present; legal standing — a protocol that pays "tax" to credential holders has no name yet in any jurisdiction.

One more thing must be said: there is a distance between design and deployment. Three layers are the design; what runs on-chain is an implementation, and an implementation need not cover the whole design. Readers should judge by on-chain facts, not by text. That applies to every protocol, including this one.

Measured by Part 2's ruler:

Basis independent of labor — holds in design. Its basis is existence in time.

Capture and allocation separated — holds in design. The tax does not pass through a capturer's hands; it goes directly to credential holders. But "in design" must be read with emphasis: if the rate is set by a few, allocation returns to some capturer's hands.

The three tests of consent — visibility holds; on-chain transfers are public. Zero-cost refusal partly holds; exit is easy, but entry requires first proving existence, and proof has a cost. Time as a ledger variable holds — this is its most fundamental difference from other schemes.

Verifiability — holds. Everything on-chain can be inspected by anyone.

Of the four marks, it passes all four in design and only "verifiable" in operation; the other three stop at "holds in design." That is its present position: a solution that can be verified, not a solution that has been proven.

What could kill it

Since the standpoint is third-party, list the ways it could fail, in my order of likelihood.

The Sybil economics flip. Once the cost of faking an existence falls below the share that existence collects, faking turns from a loss into a profit, and the ledger would be flooded with false existences within weeks. Machines are making "a convincing fake presence" cheaper every year; that line approaches annually.

Capture of the rate. Who sets the layer-three rate is the thinnest point of the whole design. If a few holders decide, they will move the rate in their own favor; if the majority decides, the majority will vote itself a raise — both outcomes return allocation to some capturer's hands, and separation fails. No known governance mechanism avoids both at once.

No buyers. If layer two's market has only sellers for long enough, the credential is just a record with no path to redemption. The record has value in itself — it is time's name on the ledger — but a name that cannot be redeemed does not retain users.

Legal classification. An on-chain protocol that pays "tax" to credential holders will, in most jurisdictions, be classified as a security, a payment instrument, or a lottery, and any of the three would bar it from operating lawfully there. The protocol cannot solve this itself; it depends on people outside it.

The design–deployment gap. How many of the three layers are on-chain and how many remain in documents can only be checked on the chain, not learned from anyone's word.

Listing these five is not an attempt to talk the thing down. A scheme whose failure modes can be enumerated is more credible than one whose cannot — the former is a mechanism, the latter a faith.

How to check a protocol

A method, in passing, that applies to any on-chain protocol claiming to solve distribution, not only this one. Open a block explorer and check five things. The contract's creation date — it decides whether "running for N years" is true. Holder and transfer counts — they decide whether "used by N people" is true. Whether the source is verified — it decides whether you can see what it actually does, rather than what it says it does. Who holds the

admin keys — it decides whether separation is written into the code or into a promise; a protocol whose tax rate can be changed by a single key has separation only in its documentation. Where the tokens flow — it decides whether the "tax" really reaches credential holders or a handful of addresses. Check all five, and you will know more about the protocol than any article about it can tell you, including this one.

Why there should be many

I said in Part 1 that the three thefts are not the work of three villains but a gap in the ledger. Closing the gap should not be the work of one scheme either.

The reason is the separation principle itself. If the world had a single protocol of distribution, it would be the sole allocator, and a sole allocator sooner or later becomes a capturer — precisely what separation exists to prevent. Competition among protocols of distribution is itself the guarantee of separation: with no single allocator, no one can enclose.

So this part ends not with a recommendation but an invitation. If you think one of the three layers is wrong, write your own version, measure it with the same ruler, and put it on a chain. The criteria are already here: a basis independent of labor, capture separated from allocation, the three tests of consent, verifiability. A scheme need not be perfect on all four, but it must answer all four.

Part 4 leaves the protocol and returns to the person. A ledger can record time, but what produces time is still a body that will die.

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Fiat steals your past, algorithms steal your present, AI steals your future. Time sovereignty is taking all three back.

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Gradient

The Theft of Time • Part 4 • Expanding Section III of the Declaration

> *Entropy is the universe's tax. Order is what you buy with a gradient. Your body, your attention and your tribe are the only three gradients you have. — The Sovereignty of Time*

Life does not fight entropy

The first three parts were about ledgers. This one is about where the thing the ledger records comes from: time comes out of a body that will die, and a body is a thermodynamic machine.

The second law of thermodynamics says that the entropy of an isolated system never decreases — order always decays, hot things cool, tidy things scatter. The sentence has been quoted so often that it has spawned a popular melancholy: if all order is doomed, all effort is a futile struggle against the universe. I wrote that way in an earlier draft a year ago. It was wrong, and harmfully so, because it asks the wrong question.

In 1944, Schrödinger asked the right one in *What Is Life?* Life is not an isolated system; it is an open one. It does not fight entropy; it "feeds on negative entropy" — it takes in order from its environment and expels the entropy it produces. Thirty years later Prigogine turned this into physics: in open systems far from equilibrium, as long as a gradient is maintained — of temperature, concentration, energy — order arises spontaneously. He called such structures dissipative. A pot of water heated from below spontaneously organizes into regular convection cells; a hurricane is the atmosphere's vast self-organization over a temperature difference; a cell is a dissipative structure over chemical gradients. Order is not won by fighting entropy; it is bought by consuming a gradient. Cut the gradient and the order collapses at once.

So the right question is not "how do I fight entropy." Entropy is the universe's tax; no one is exempt. The right questions are two: What is your gradient? And is someone else collecting it?

Look back at Part 1: the three thefts take exactly three gradients. Fiat drains the reserves your body earned; algorithms shred your attention; AI makes your expertise no longer enough to gather a tribe around. This part takes the three in turn — they are the only three you have.

Body: energy becomes time

The body is a machine that converts energy into usable hours. Its efficiency is how many waking, focused, pain-free hours each unit of energy buys. That definition compresses every discussion of health into one question: does this increase my gradient, or consume it?

On the evidence about the body, the honest approach is to grade it, because the field is full of exaggeration.

The evidence for strength training is the firmest. In 2022, Momma and colleagues published in the *British Journal of Sports Medicine* a meta-analysis of sixteen prospective studies finding that 30 to 60 minutes of muscle-strengthening activity a week was associated with a 10 to 17 percent lower risk of all-cause mortality, with corresponding reductions in cardiovascular disease, cancer and diabetes. This is observational evidence, not proof of causation, but its consistency and dose-response are clear. Muscle is the body's largest metabolic organ; its stock sets your energy-conversion efficiency, and it is lost at several percent a decade from the age of thirty onward — untrained, the gradient drains at a steady rate.

The mechanistic evidence on sleep comes from Xie and colleagues in *Science* in 2013: during sleep, the clearance of metabolic waste from the brain by cerebrospinal fluid rises markedly. The experiments were in mice and extrapolation to humans needs care, but they gave "sleep cleans the brain" a physical mechanism for the first time rather than a metaphor. The brain is the most expensive dissipative structure you own; if its waste is not cleared, its order cannot be rebuilt.

The evidence on intermittent fasting is in the middle. De Cabo and Mattson's 2019 review in the *New England Journal of Medicine* summarized the animal and human trials: in animals, the evidence that time-restricted feeding and caloric restriction extend life is strong; in humans, the evidence concentrates on metabolic markers — glucose, lipids, weight — and evidence on lifespan does not yet exist. Ohsumi won the 2016 Nobel Prize for elucidating autophagy, which is the mechanistic foundation, but "fast for sixteen hours to activate autophagy and slow aging" skips several steps between mechanism and conclusion. In my earlier draft I labeled this "entropy reduction at the cellular level." That was labeling, not argument.

So this section gives no checklist. It gives a principle, borrowed from Taleb: organisms grow stronger under moderate stress and weaker under sustained comfort — what he calls antifragility. The body's gradient is not maintained by reducing consumption but by a cycle of stimulus and recovery: training is stimulus, sleep is recovery; hunger is stimulus, eating is

recovery. Comfort removes the stimulus and thereby the cycle. "Comfort is the hotbed of entropy; stress is the trigger of order" means, physically, that a gradient must be rebuilt repeatedly, not protected.

Body fat percentage is the body's inflation rate — not an aesthetic figure but the speed at which reserves are being diluted. That is one line from the old draft I am keeping.

Attention: time becomes order

The body turns energy into hours; attention decides what those hours become. An hour without attention is an hour spent and exchanged for no order.

In 2010, Killingsworth and Gilbert published in *Science* a study that pinged more than two thousand people at random moments through their phones, asking what they were doing, what they were thinking about, and how they felt. The result: for 46.9 percent of waking hours, people were thinking about something other than what they were doing; and while their minds wandered — whether toward pleasant or unpleasant things — they were less happy. The study matters because it ties attention to experience: the integrity of attention determines not only what you produce but how you live. A fragmented present is not merely inefficient; it is unhappy.

Part 1 cited Mark's 2008 study: after an interruption, returning to the original task takes about twenty-three minutes, and people compensate by speeding up, at the cost of stress. Together the two studies draw the portrait of a person whose gradient has been collected by the algorithm: half the time elsewhere, unhappy while elsewhere, twenty-three minutes to come back each time, and more tense for the effort.

Newport calls the opposite capacity deep work: long, uninterrupted stretches of pushing one's cognition to its limit. His argument is that this capacity is becoming scarcer and more valuable at once — scarcer because the environment systematically destroys it, more valuable because it is ever harder to replicate. Part 1 argued that the last thing to be priced in the age of AI is judgment; judgment arises only in deep attention.

Deep attention also has a ceiling, and knowing it matters. Ericsson's 1993 study of violin students at the Berlin conservatory found that the very best practiced deliberately for about four hours a day, in several sessions with rest between them; beyond that, the quality of practice fell and the gains disappeared. Later studies in other fields converged on similar figures. The implication: a person can genuinely turn time into order for only three or four hours a day; the remaining hours are maintenance, recovery and expenditure. That makes the attention gradient precise — you have about four hours a day of time with a name on it, and those four hours are exactly what the algorithm wants, because only those four are worth auctioning.

In practice, the attention gradient is kept by four things. Time blocks: cut the day into a few large pieces, one task per piece, every notification off inside it. Environment: a physical place used only for focus — a room, a desk, even a pair of headphones — so the body knows that entering here means entering focus. Single-tasking: parallelism is an illusion; the brain has no parallel capacity, only fast switching, and every switch costs twenty-three minutes. External memory: hand memory to a note system — Forte calls it a "second brain" — whose purpose is not to remember more but to free attention from "remembering" for "judging."

Attention sovereignty has a simple test: can you go ninety minutes without looking at your phone? If not, the gradient has already been collected, and whoever collected it did not ask.

Tribe: order compounds

One person's order dies with them. A group's order can be passed on — language, craft, trust, rules are all order that compounds between people. This is the third gradient, and the easiest to overlook, because it does not look like thermodynamics.

In 1992, Dunbar extrapolated from primate brain size the upper limit on the number of stable social relationships a human can maintain: about one hundred and fifty. The number does not mean "you should know a hundred and fifty people." It means tribes are small by nature. The capacity for deep connection is limited, therefore scarce, therefore a gradient. Ten people you can call at three in the morning are closer to a tribe than a thousand who have liked a post.

Why is the tribe the most important of the three? Because it is the only one a machine cannot copy at zero marginal cost. Trust is not content that can be generated; it is the result of repeated play. Axelrod showed in 1984, with a computer tournament, that cooperation evolves in repeated games — provided the game repeats often enough and the players remember the last round. Trust requires presence over time: the same person, at many moments, doing what could be expected. That is precisely what machines lack. Machines have compute but no duration — they are not "present" in time, so they cannot be trusted, only used.

Part 1 said AI makes expertise no longer enough to gather a tribe. The answer is here: what gathers a tribe was never expertise but presence. Expertise was only one way of proving presence, and as that proof depreciates, the value of presence itself comes into view. This is the same thing as the first layer of the protocol in Part 3, at a different level — the chain records "was here in time," and so does the tribe.

Dunbar later divided the hundred and fifty into layers: five intimates, fifteen you can call on, fifty friends, a hundred and fifty acquaintances. Each layer outward, the time each person receives drops by an order of magnitude. The gradient is in the innermost two layers — five

and fifteen — because only there is the density of presence enough for trust to compound. The outer hundred and thirty-five are environment, not gradient.

Tribes have their own failure mode, which must be said. A tribe that only agrees with you is not a gradient but a drain — it deprives your judgment of correction, and a person whose judgment loses correction is losing order just as surely as one whose attention is being shredded. The test is simple: in the past year, has this group changed your mind about anything? If not, you have an echo chamber, not a tribe.

In practice, the tribe gradient is kept by three things. Build in public: make what you are doing, your judgments and your failures visible; this is the most verifiable form of presence, and it lets the right people find you. Give value first: in any relationship, ask what you can give before what you can take, because the first step of trust is always someone paying first. Depth over breadth: spend time on ten people, not a thousand.

Guarding the gradient is thermodynamics

Compress this part into a table. The body turns energy into time; fiat drains the reserves it earns. Attention turns time into order; algorithms shred and auction it. The tribe compounds order; AI removes the reason for gathering it. Three gradients, three thefts, one to one.

Guarding the gradient is not self-improvement advice; it is thermodynamics. Without a gradient there is no order; without order there is no time to record. The ledgers of the first three parts record time; the body, attention and tribe of this part are where time comes from. If the source dries up, the ledger is empty.

So keep a ledger of your own — three numbers a week are enough. How many strength sessions this week. The longest uninterrupted stretch of focus this week. How many concrete things you did this week for those five or fifteen people. If the three numbers fall for several weeks running, you need no one to tell you: your gradient is being collected — and the collector is usually one of the three systems from Part 1.

But gradients run out in the end. The body dies, attention goes dark, the tribe disperses. Where does the order go then? That is the question for the last part.

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Fiat steals your past, algorithms steal your present, AI steals your future. Time sovereignty is taking all three back.

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Endgame

The Theft of Time • Part 5 • Expanding Section VI of the Declaration

> *The endgame is not escape. It is continuation. The end of the body is set by biology. The end of influence is set by the ledger. — The Sovereignty of Time*

The escape narrative, and where it ends

Books about the "sovereign individual" almost all stop in the same place: escape.

In 1997, Davidson and Rees-Mogg predicted in *The Sovereign Individual* that information technology would weaken the nation-state's hold on the individual; the capable would no longer be locked to geography — income from everywhere, assets encrypted, the state demoted from master to a service provider one may choose. In 2022, Srinivasan's *The Network State* pushed the prediction a step further: first gather an online community around shared values, then crowdfund territory, then seek diplomatic recognition — network first, then state.

The insight in both books is real. States are indeed becoming optional; borders are indeed thinning. But both books answer the same question: where are you. Neither answers the other question: where is your time going. Moving does not solve time — in whatever jurisdiction you place yourself, time runs out at the same speed, and fiat, algorithms and models operate identically in every one of them. Changing your passport changes who collects your tax; it does not change the ledger.

In an earlier draft, a year ago, I wrote advice such as "obtain a second passport." This part corrects it: that was an answer about geography, and the problem is not geographic.

Hirschman, in 1970, gave a more general frame. Faced with an organization that disappoints you, you have two responses: exit, or voice. Exit is voting with your feet; voice is trying to change it from within. The sovereign-individual narrative is pure exit; politics is voice. Time sovereignty takes a third road: neither exit nor voice, but a different ledger. You do not leave the system and you do not try to persuade it; you build, beside it, a place that records time, and let the record speak. The protocol of the first three parts is one instance of that third road.

So the endgame is not escape. Escape is a spatial act, and the endgame of time sovereignty has to be temporal: continuation.

The end of influence is set by the ledger

The end of the body is set by biology, and for now no one can change that. But whether a person's thought, judgment and work keep acting after death — that is not set by biology. It is set by where they are recorded.

Where you record decides how long it lives. Consider three ways of recording.

Recorded on a platform, it vanishes with the platform. The Pew Research Center found in 2024 that 38 percent of webpages that existed in 2013 were no longer accessible by 2023; pages on government and news sites were disappearing at similar rates. A 2021 study by the Harvard Law School Library examined deep links inside *New York Times* articles and found a quarter of them dead. These are not edge cases; they are the internet's normal condition. Every closed blogging platform, every shut-down social network, every redesigned site takes a cohort's records with it, and the people taken were never notified. The internet promised memory and delivered forgetting.

Recorded on a hard drive, it dies with the drive. No statistics are needed; everyone has lost at least one.

Recorded in the memory of descendants, it reaches zero within three generations. Anyone who has traced a family tree knows the figure: at the great-grandparents' generation, most people cannot even complete the names.

So "the end of influence is set by the ledger" is not rhetoric. It is a statement about storage media: your order lasts exactly as long as the medium it sits on, and all three media above have lifespans shorter than the time a person wishes to be remembered.

Some have tried designing for longer. In 1996, Brand and Hillis founded the Long Now Foundation to build a clock that runs for ten thousand years and chimes once a century. The point is not the clock but the question it forces: if you had to design for ten thousand years, how would you do it? The answer is nothing like "design for ten years" — you cannot rely on any company, any power source, or any one generation's memory. That is the correct scale for the archive problem: not "how do I preserve this past my death" but "how do I preserve this past the point where no one remembers who I was."

An archive that never closes

Is there a medium whose preservation depends on no institution's goodwill?

In 2019, Williams and colleagues described in the Arweave yellow paper a mechanism: pay once, store permanently. Its economic model is an endowment — of the fee you pay, a small part covers storage now and most goes into a fund whose returns pay for storage in the future. The model rests on one assumption: that the cost of storage keeps falling. Historically it has fallen about 30 percent a year; the model uses a far more conservative rate and arrives at a floor of at least two hundred years. Nodes keep the data not out of goodwill but incentive: a node must prove it can access randomly chosen historical data in order to earn the right to write new blocks.

The limits of this scheme must be stated plainly. Two hundred years is an economic assumption, not a physical guarantee; the network could die; the incentives could fail. But it differs from the three media above in one essential way: it is at present the only scheme that has converted "preservation" from an institution's goodwill into an economic incentive. A platform keeps your things because they are useful for now; a hard drive keeps them because it has not yet failed; a permanent archive keeps them because someone profits from doing so, and the structure of that profit was designed so that no one has to decide. That is what "kept alive by mathematics and economic incentive, not by any institution's goodwill" means.

For an individual, this means three things. Your thought, decisions and work can be carved into a place that does not vanish with a platform. What is carved cannot be altered — not because no one wants to, but because the cost of altering exceeds anyone's motive. And it depends on no one's favor — there is no account to be banned and no company to go bankrupt.

This declaration and these five essays use a lighter version of the same idea: the fingerprint of every text is recorded on the Bitcoin blockchain, and the text itself is hosted in a public paper repository. Not because they are important, but because this is what Section VI must do before it can ask anyone else to.

Digital inheritance

The archive answers "where to record." Inheritance answers "how what is recorded keeps acting." Three things.

First, inheritance of the existence credential. Weyl, Ohlhaver and Buterin, in a 2022 paper, described a class of non-transferable credentials — soulbound tokens — bound to one person, unsellable, but able to carry rules for recovery and inheritance: a person you designate, under conditions you set, takes it over. The existence credential of Part 3 is a credential of this kind. Its inheritance means your column on the ledger does not disappear with you.

Second, automatic settlement of the ledger. A smart contract can go on executing the rules you set when you are no longer present — allocating, transferring, paying according to your wishes, triggered by a signal you no longer answer. This is not new technology; it is the on-chain version of the dead man's switch: your judgment becomes code, and the code goes on settling after you.

Third, an agent of your judgment. A lifetime of a person's writing, decisions and work is enough to train a model close to them in style and inclination — one that can go on writing, evaluating, answering. This was the most radical part of chapter twelve in the old draft, and this part has to draw its boundary.

The boundary is this: the model is not you. It is the statistical shadow of your judgment — like you on the questions you wrote about, guessing on the ones you did not. It has no gradient of yours — it will not die, so it does not treasure, and treasuring is where judgment comes from. So its authority must be limited, and the limit follows directly from the separation principle: it may advise but not decide; it may allocate but not capture. An agent that can capture value on your behalf will sooner or later become an agent that captures value for itself.

Then there is the consent of descendants. A "digital ancestor who is never offline" may be a fortune to the next generation or a burden. The design must let them switch it off. Continuation is not imposition; the right to be remembered does not include the right to force others to remember.

One more thing is unsolved: the law. Estate law recognizes wills, trusts and heirs; it does not recognize smart contracts or soulbound credentials. Where an on-chain ledger that settles automatically and a will probated in court say different things, nearly every jurisdiction today will side with the will. So a digital will cannot replace a legal will; it can only stand beside it, and the two must say the same thing — the only safe arrangement at this stage. This is the twin of Part 3's "legal classification" problem: the ledger walks ahead, the law follows behind, and the distance between them is yours to bear.

Finally, a philosophical line, to prevent misreading. Digital inheritance is not immortality, not mind-uploading, and it does not answer "is that model me" — the ship of Theseus has no answer here and needs none. It answers a smaller, more practical question: can a person's judgment and work go on acting, by the rules that person set, after they die? Yes. That is enough.

In practice, digital inheritance begins with four things today. A unified identity: a decentralized identity and existence credential under your control, marking your ownership and your work — the entry point for any future inheritance. Structured records: write down your thinking, decisions and reflections; this is not only a journal but the corpus for any future agent and the basis on which descendants will understand your judgment.

Permanent archiving: put your most important work and documents into a permanent archive, starting this year, without waiting for completeness. A digital will: who inherits your credential, what authority the agent has, and under what conditions descendants may switch it off.

Not immortality, but not being zeroed out

If time is money, death is the final bankruptcy. The sentence is not pessimism; it is accounting. A person's time account closes at the moment of death, and all the order built before it — judgment, work, relationships — is zeroed out with the account unless it was recorded somewhere that continues.

Code has given us, for the first time, a chance at reorganization. Reorganization is not immortality. In a bankruptcy reorganization the debtor is gone, but the assets and the rules remain and can go on operating by those rules. Not immortality — but not being zeroed out.

The five parts close here. Part 1: three systems steal time, and they can because time has no name on the ledger. Part 2: the old ledger records only labor and holding; once machines supply time it cannot go on, and capture must be separated from allocation. Part 3: someone has tried to write "existence" into a ledger; measured with the ruler, it is a solution that can be verified, not the only one. Part 4: time comes from three gradients — body, attention, tribe — and guarding them is thermodynamics. Part 5: gradients run out in the end, but order recorded on a ledger that continues does not.

The declaration's last line is: from today, record it on your own ledger. After five parts the line should have a concrete meaning — record it on a ledger that records existence, on a medium that depends on no one's goodwill, in a place that goes on settling by your rules after you are gone.

That is your ledger.

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Fiat steals your past, algorithms steal your present, AI steals your future. Time sovereignty is taking all three back.

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时间主权宣言 · 完整版 — The Sovereignty of Time: A Declaration · Complete Edition

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